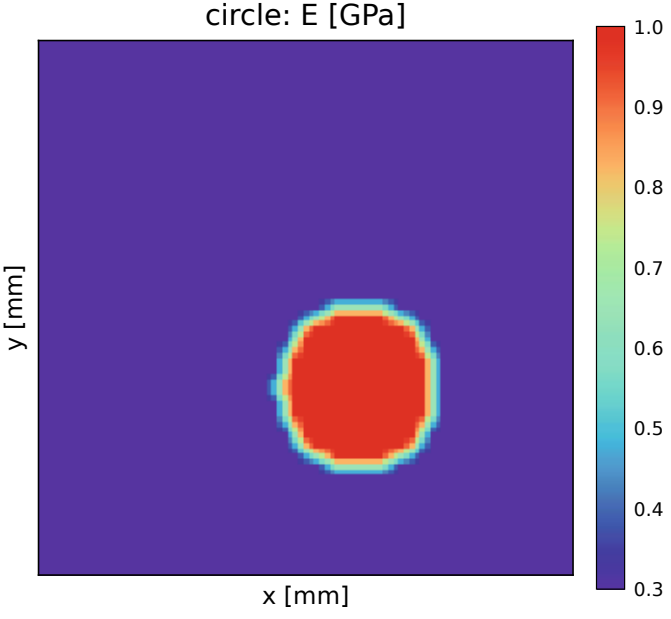
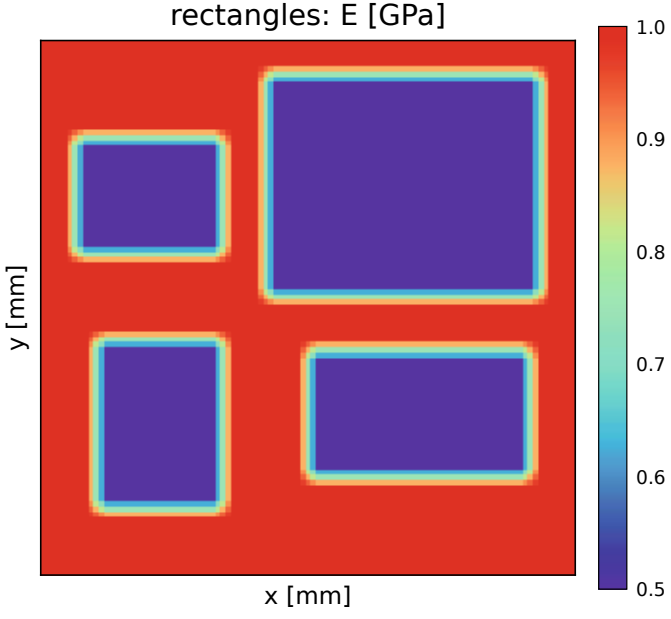




FULL CATALOGUE: verified completed jobs | Recomputed from shared geometry code; alpha=1,T0=0 and origins are reconstruction choices. Paper unit-label conflict disclosed.

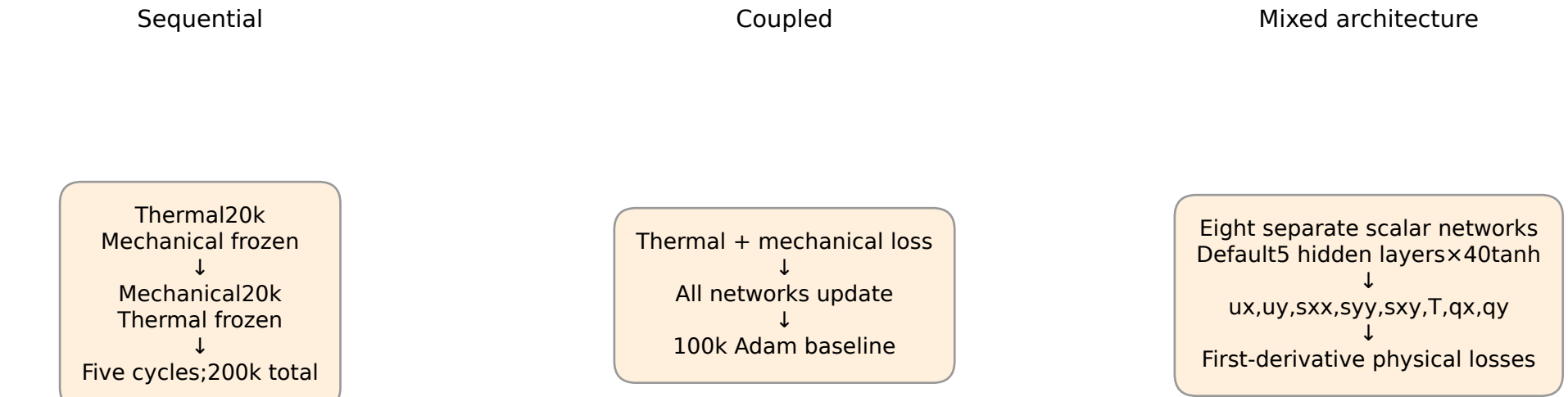
Mechanical: ux=0 at x=0,1; uy=0 at y=0,1

Thermal: T=1 left, T=0 right; qy=0 top and bottom

Soft baseline: boundary penalties in the loss

Hard cases: $u_x = x(1-x)N_{u_x}$, $u_y = y(1-y)N_{u_y}$

$T = 1 - x + x(1-x)N_T$



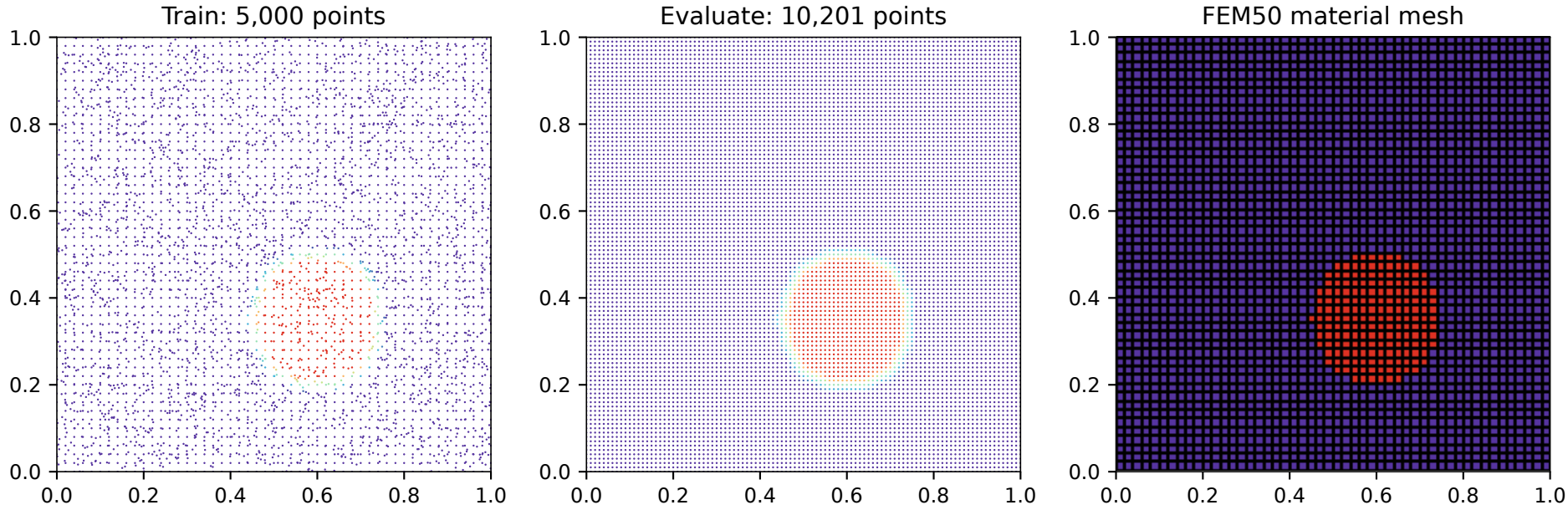
FULL CATALOGUE: verified completed jobs | New schematic. Baseline soft BCs and authors-energy variant; hard/L-BFGS and transfer cases have separate configurations.

Paired material ratios (1,1) → … → (10,10)

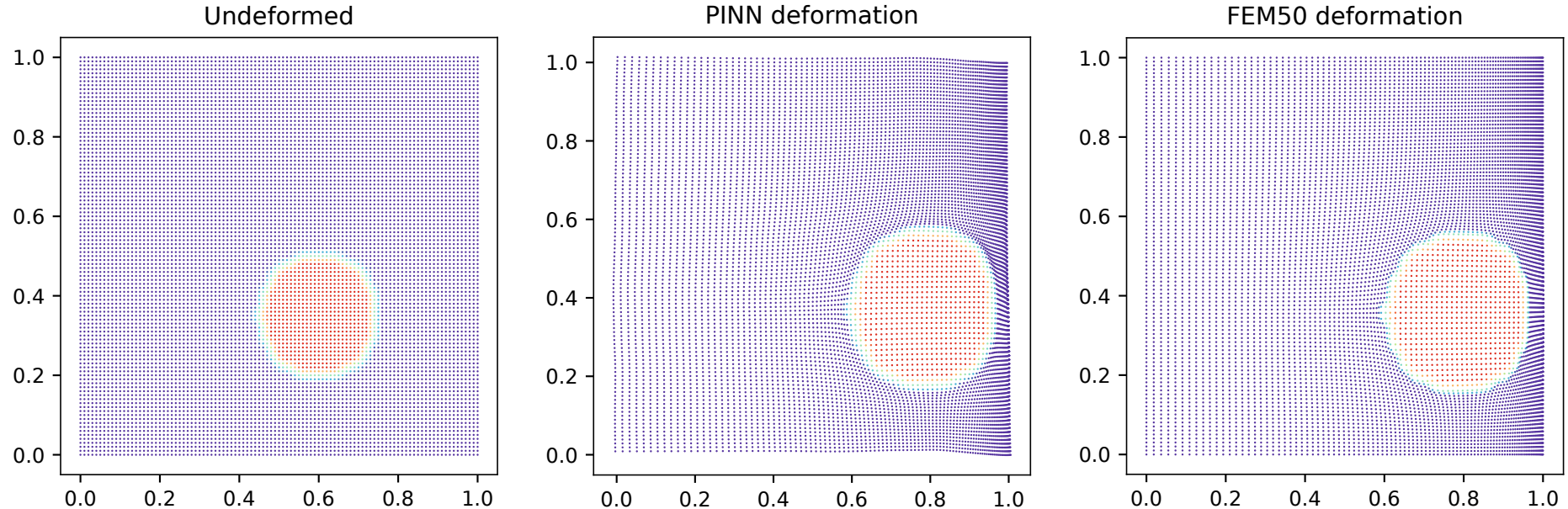
10,000 Adam updates per ratio; transfer network weights

Evaluate held-out (15,15) without further training

w = 0, 0.1, 0.5, 1; 0.5 is an explicit reconstruction choice

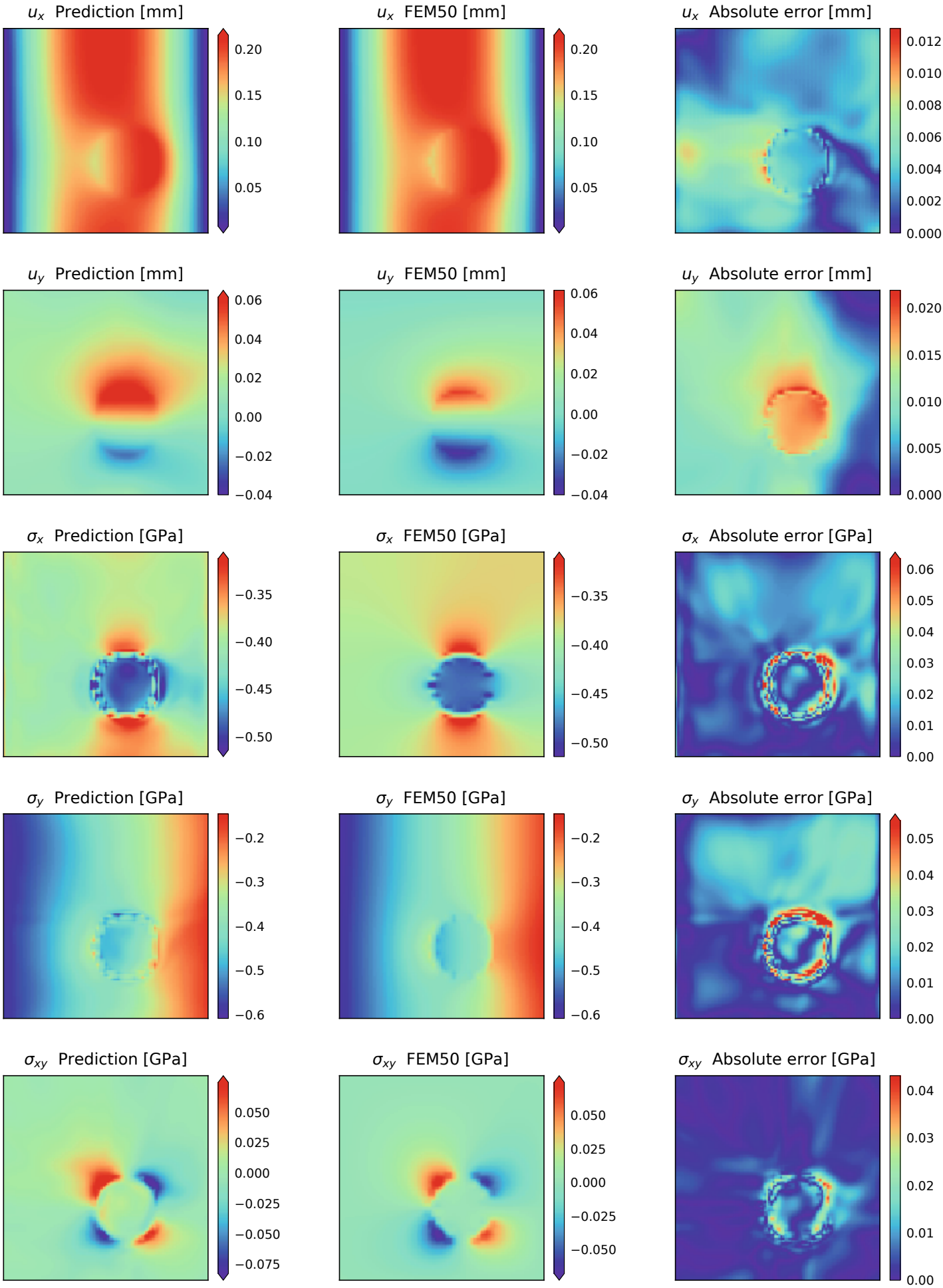


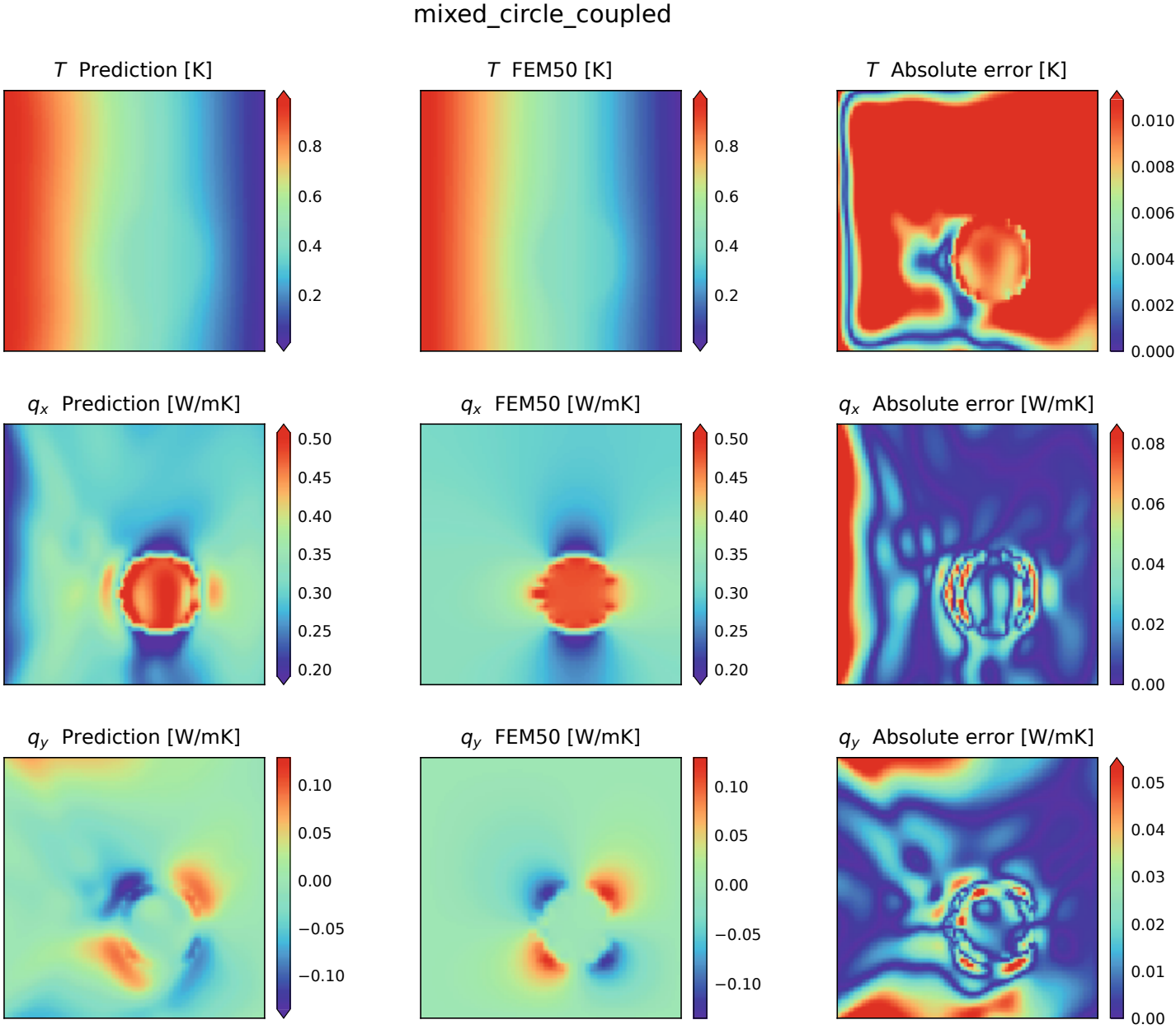
FULL CATALOGUE: verified completed jobs | Train points reconstructed deterministically from saved config and sampler; evaluation coordinates read from prediction NPZ.



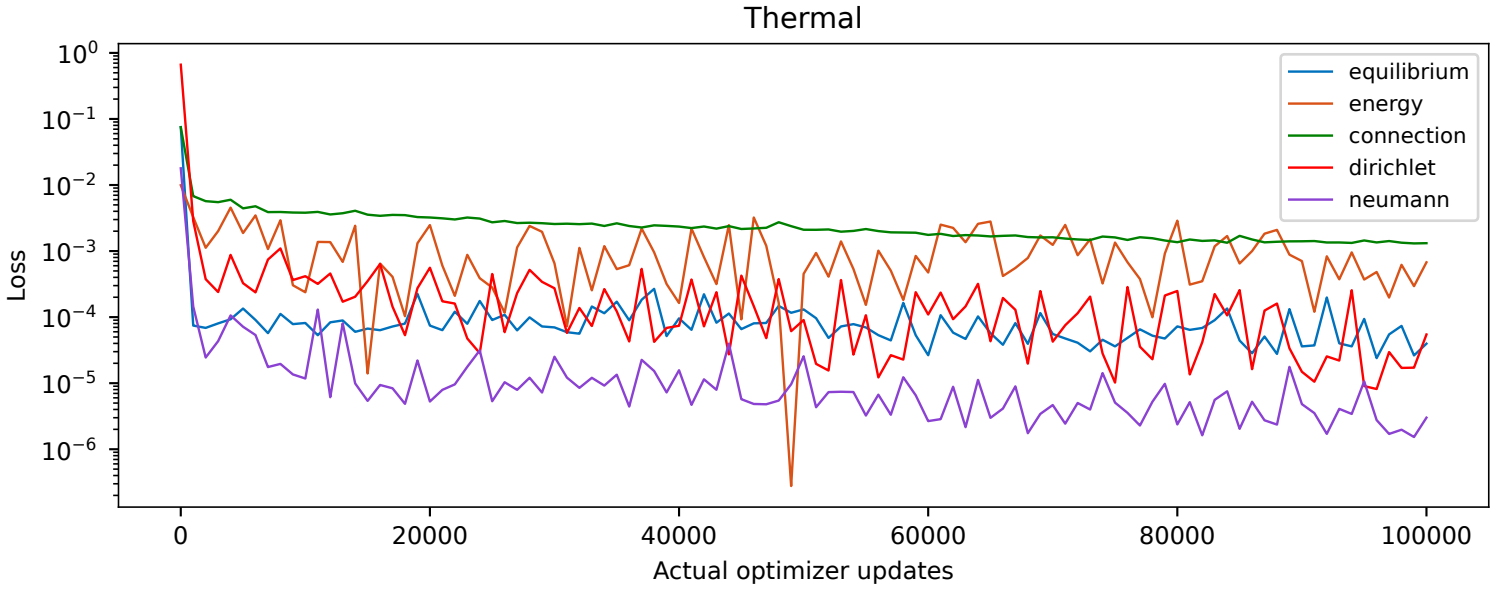
FULL CATALOGUE: verified completed jobs | Displacement factor1; no fitted amplitude or coordinate clipping. Artificial benchmark alpha=1.

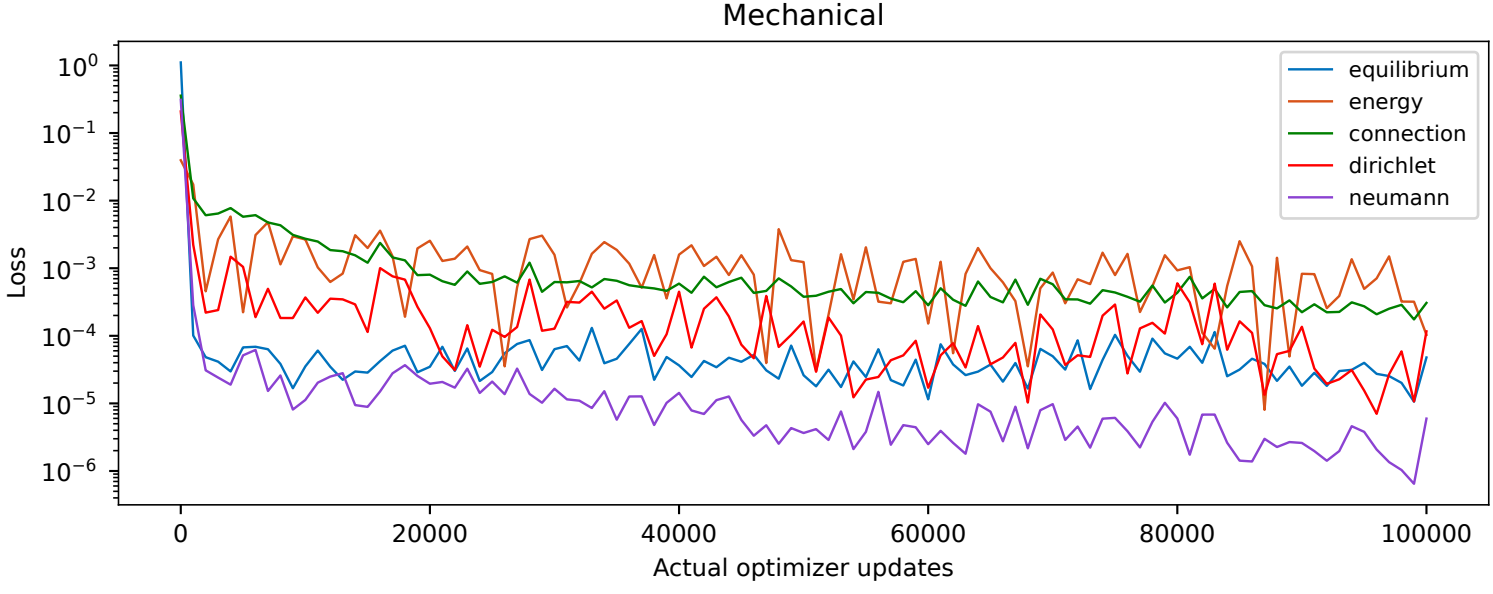
mixed_circle_coupled





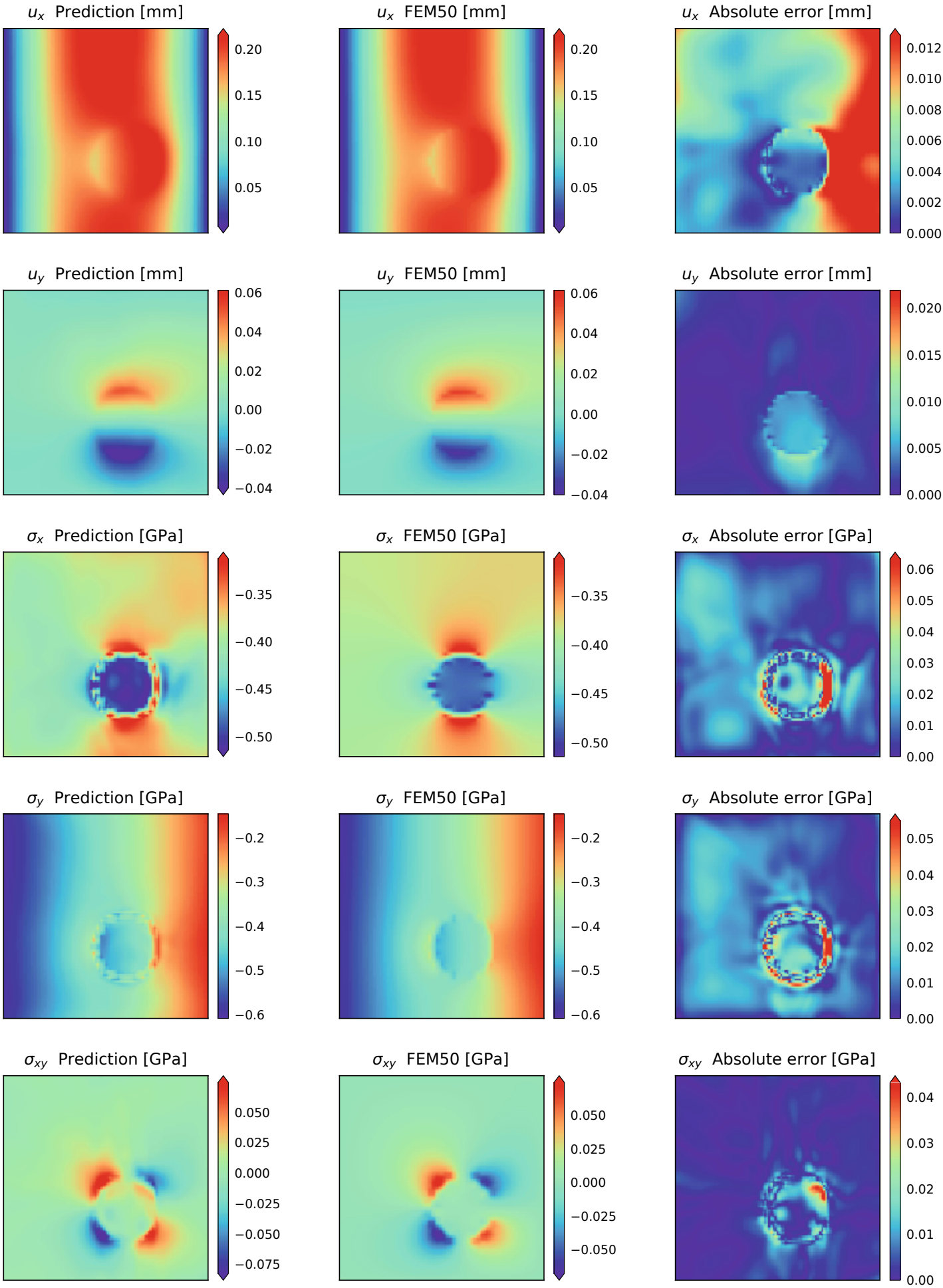
FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

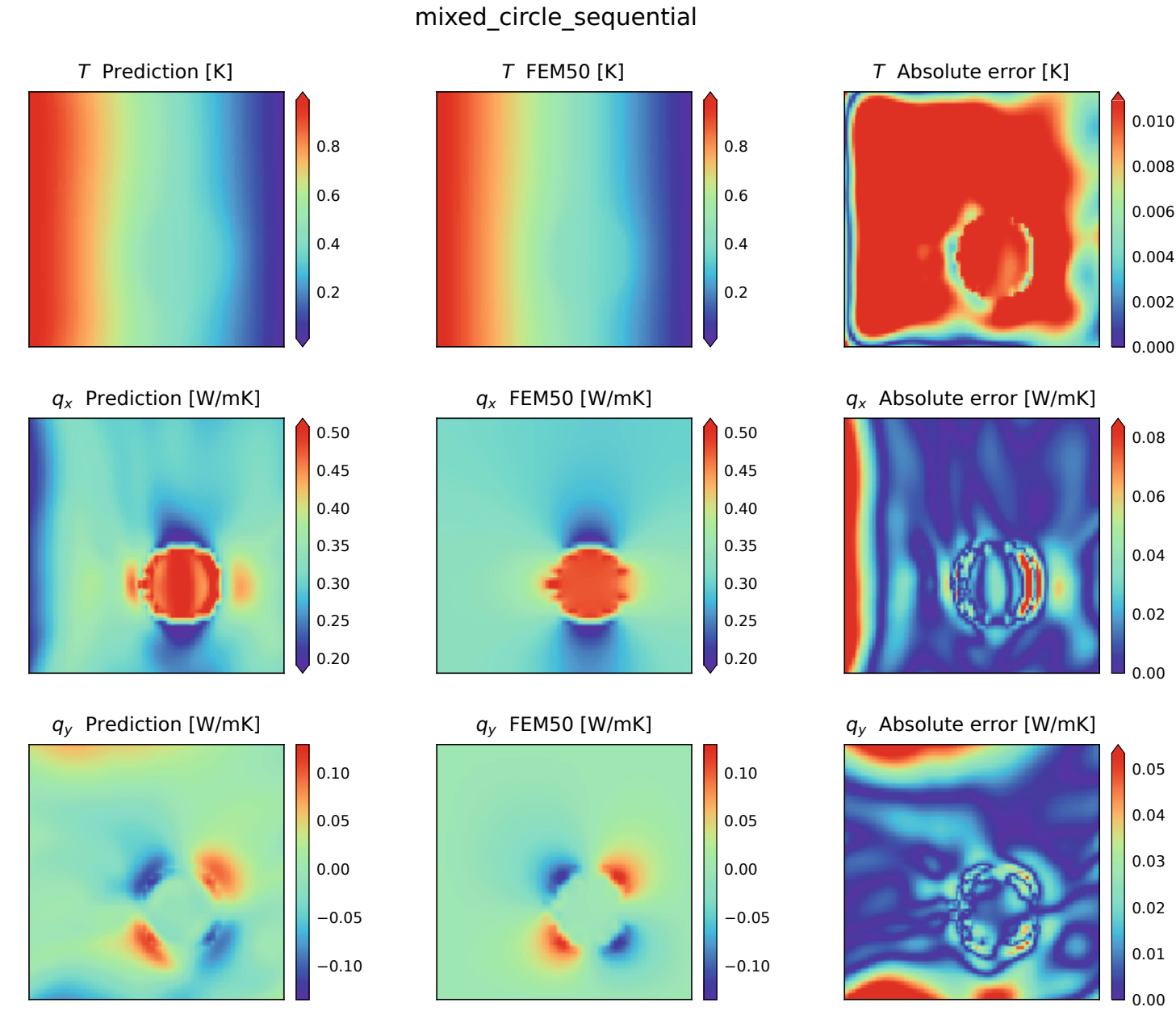




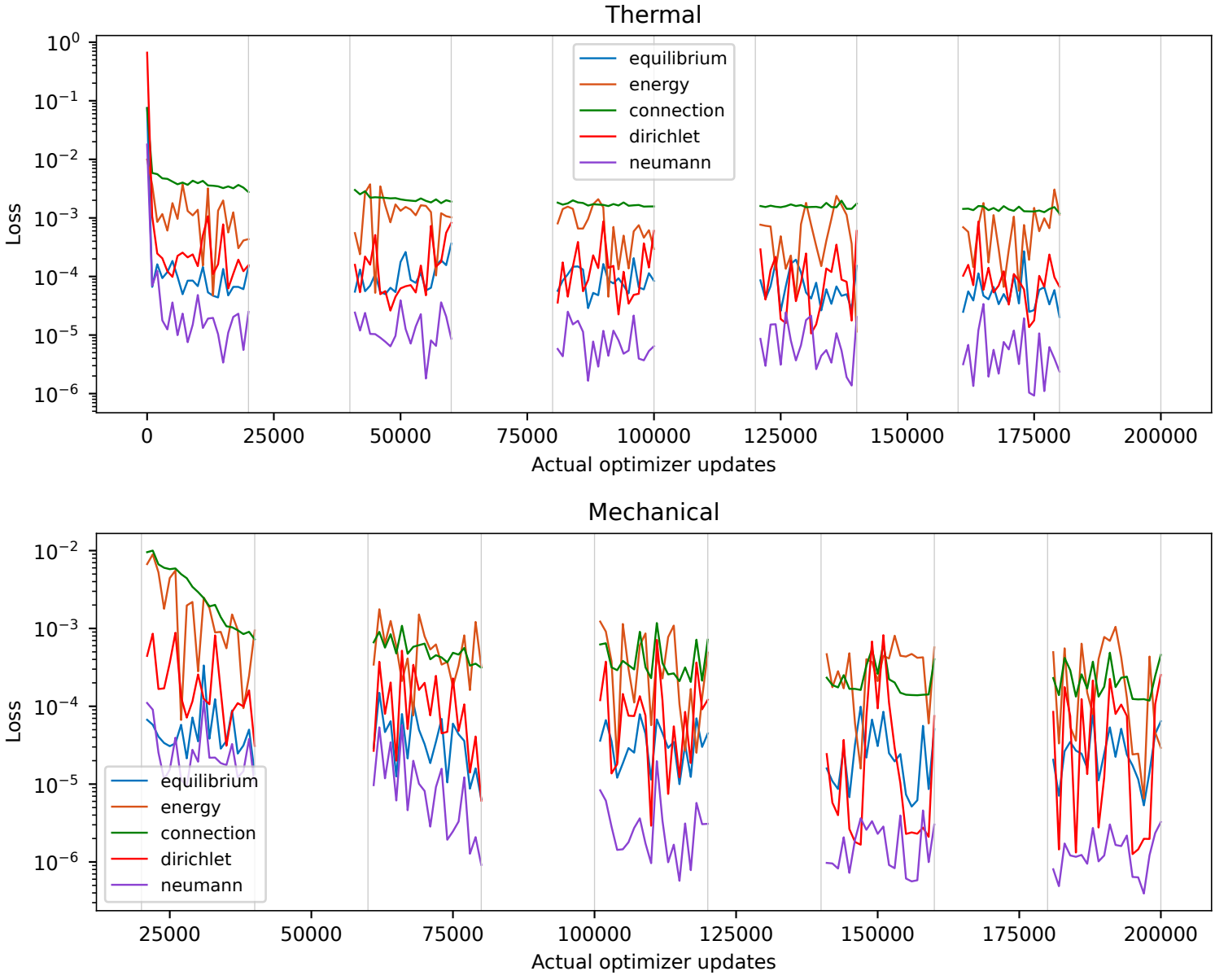
FULL CATALOGUE: verified completed jobs | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

mixed_circle_sequential

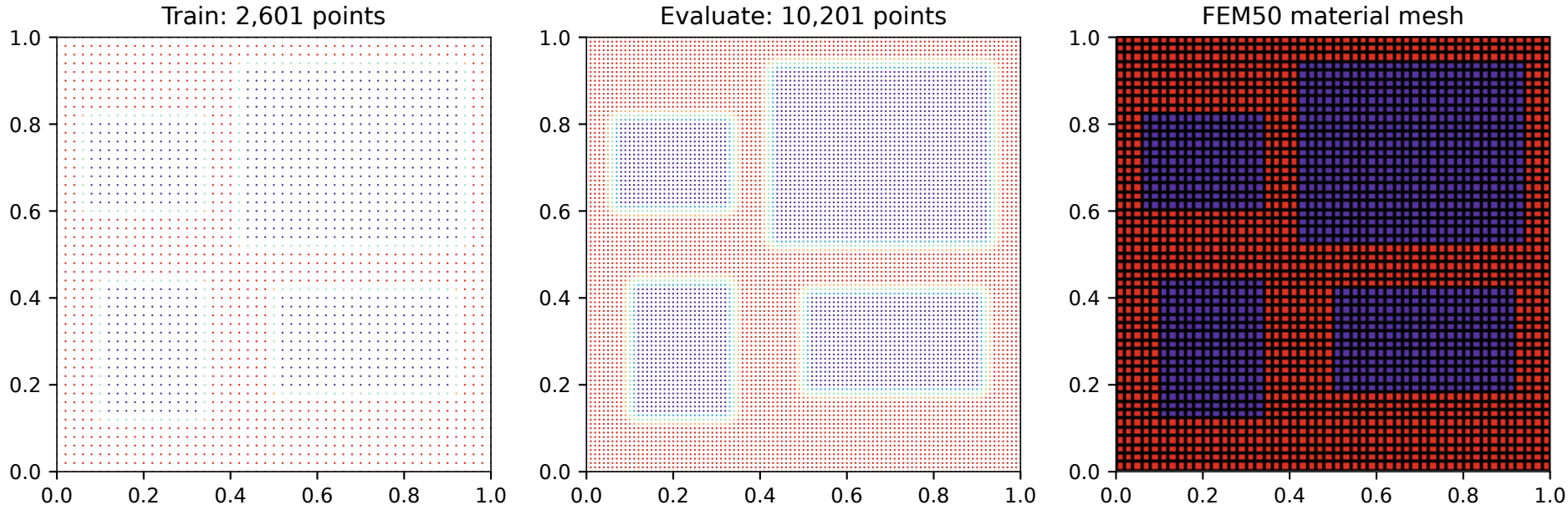




FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

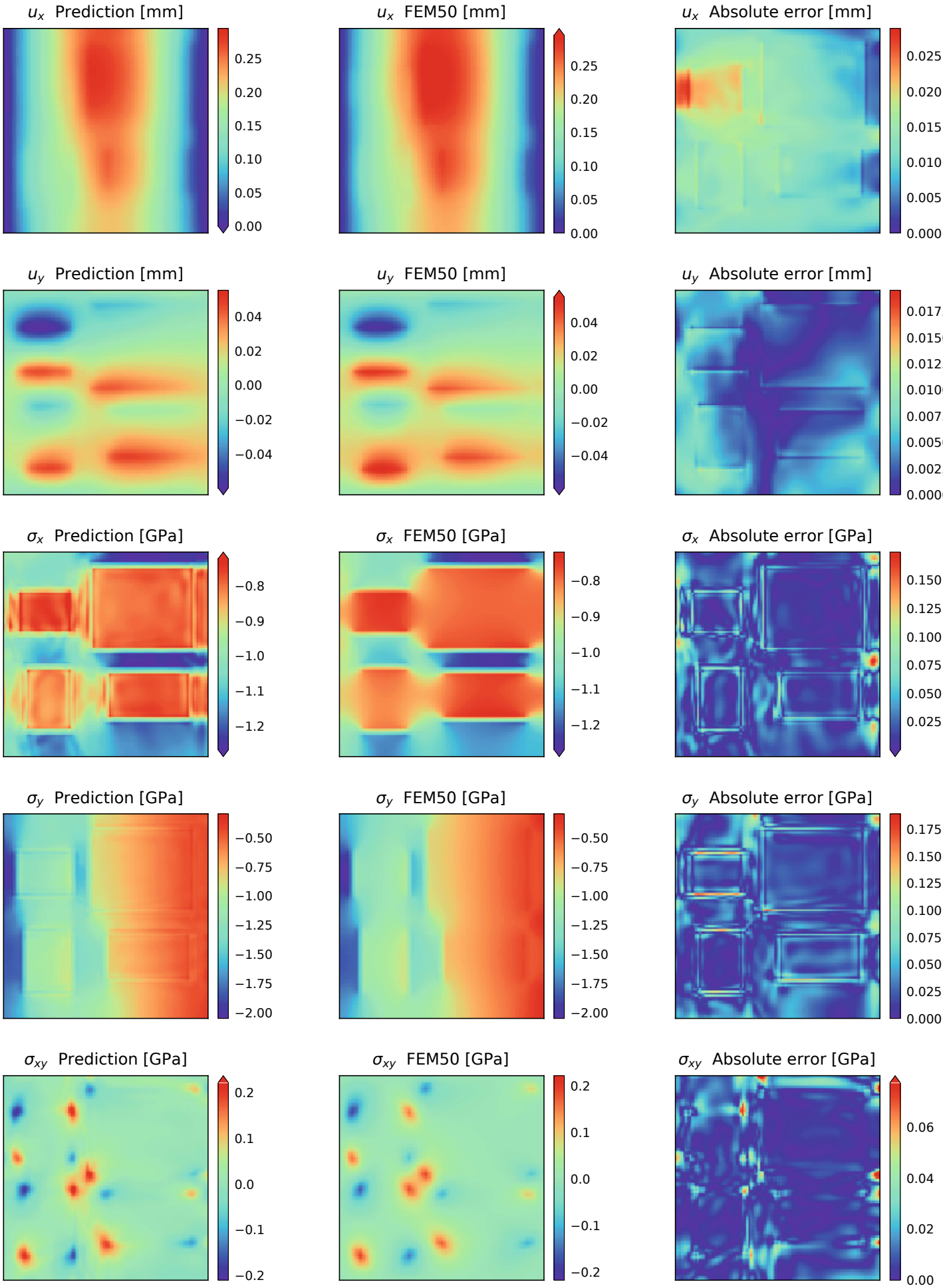


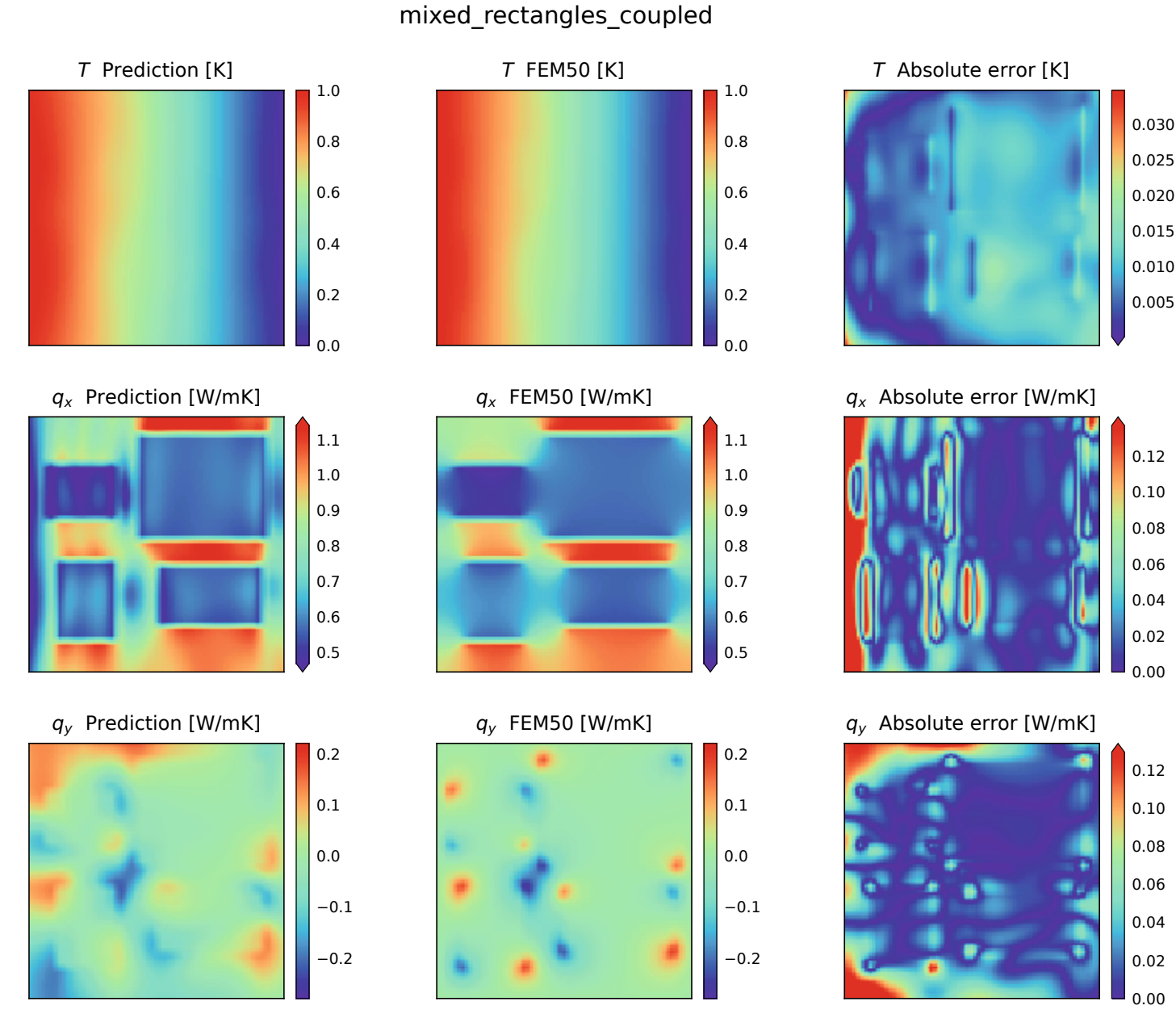
FULL CATALOGUE: verified completed jobs | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.



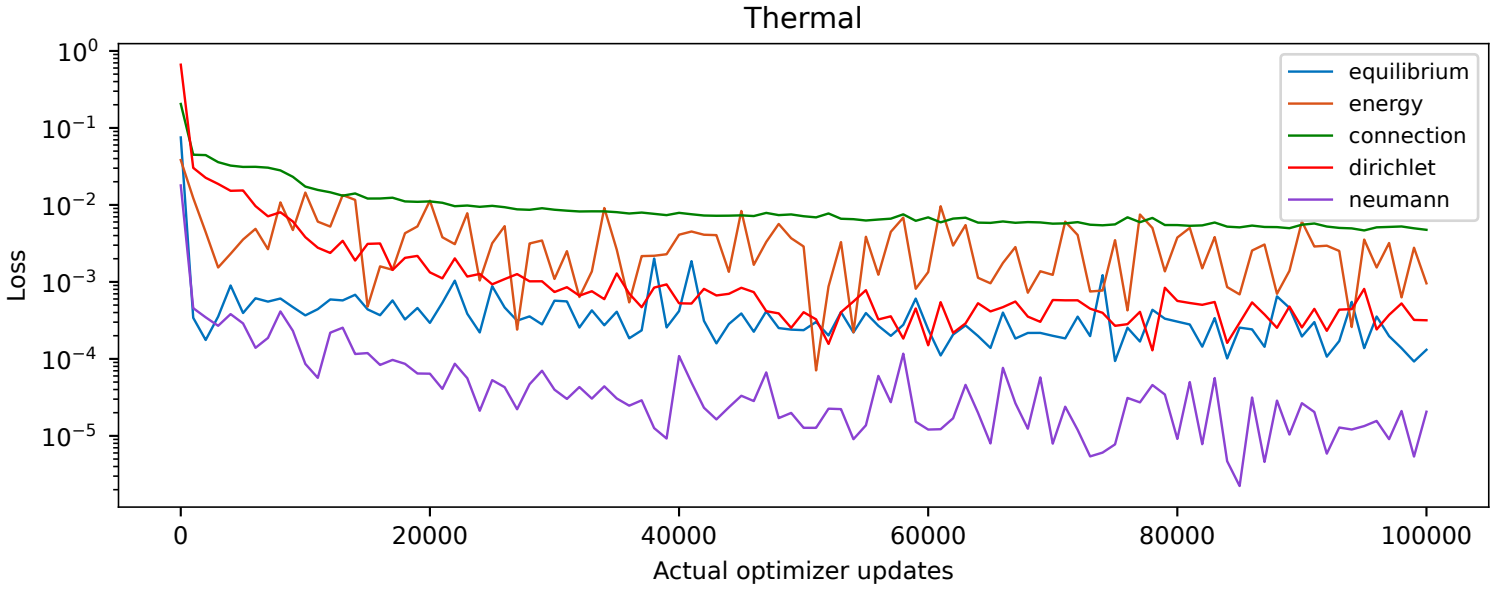
FULL CATALOGUE: verified completed jobs | Train points reconstructed deterministically from saved config and sampler; evaluation coordinates read from prediction NPZ.

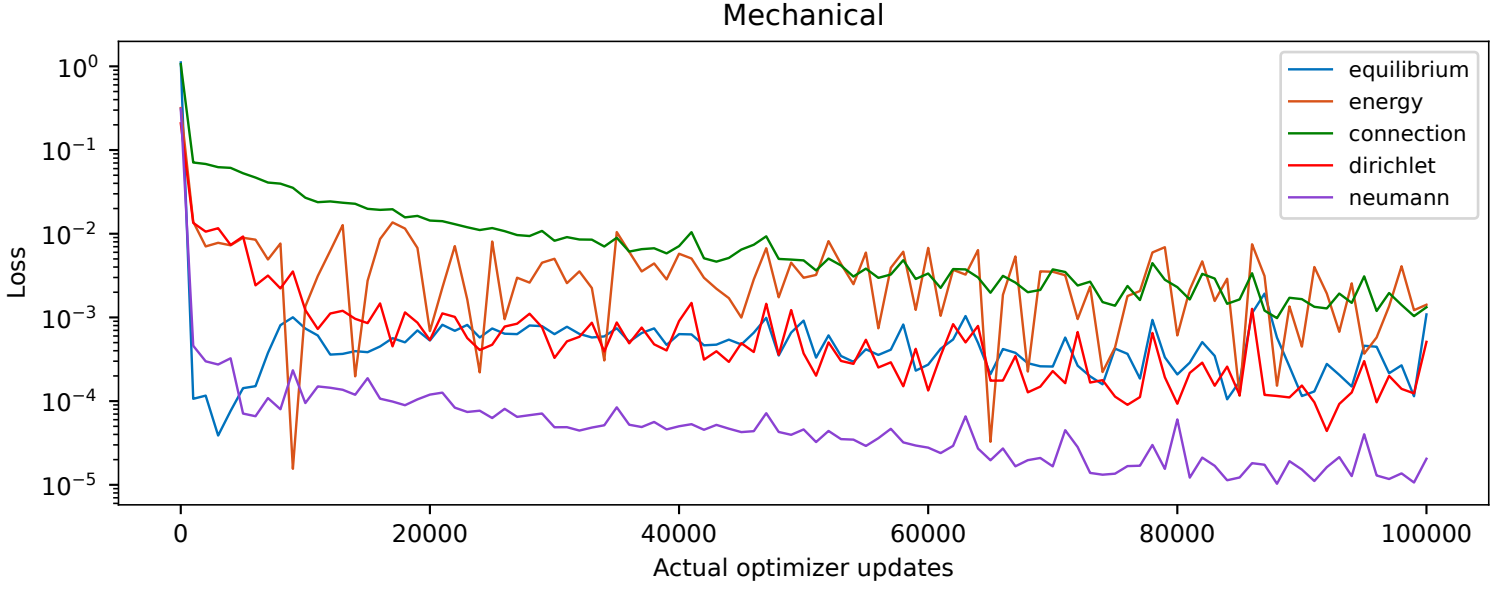
mixed_rectangles_coupled





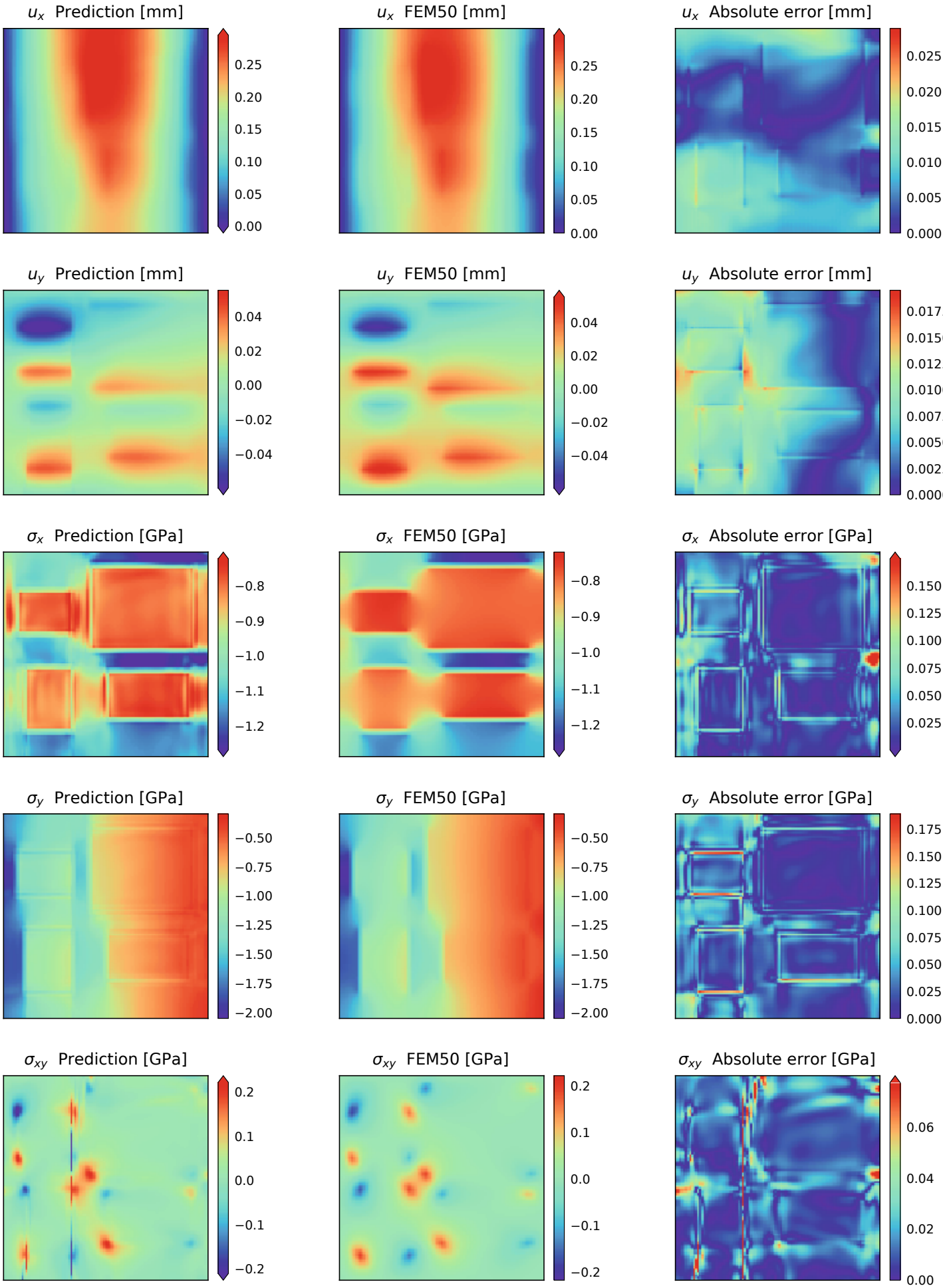
FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

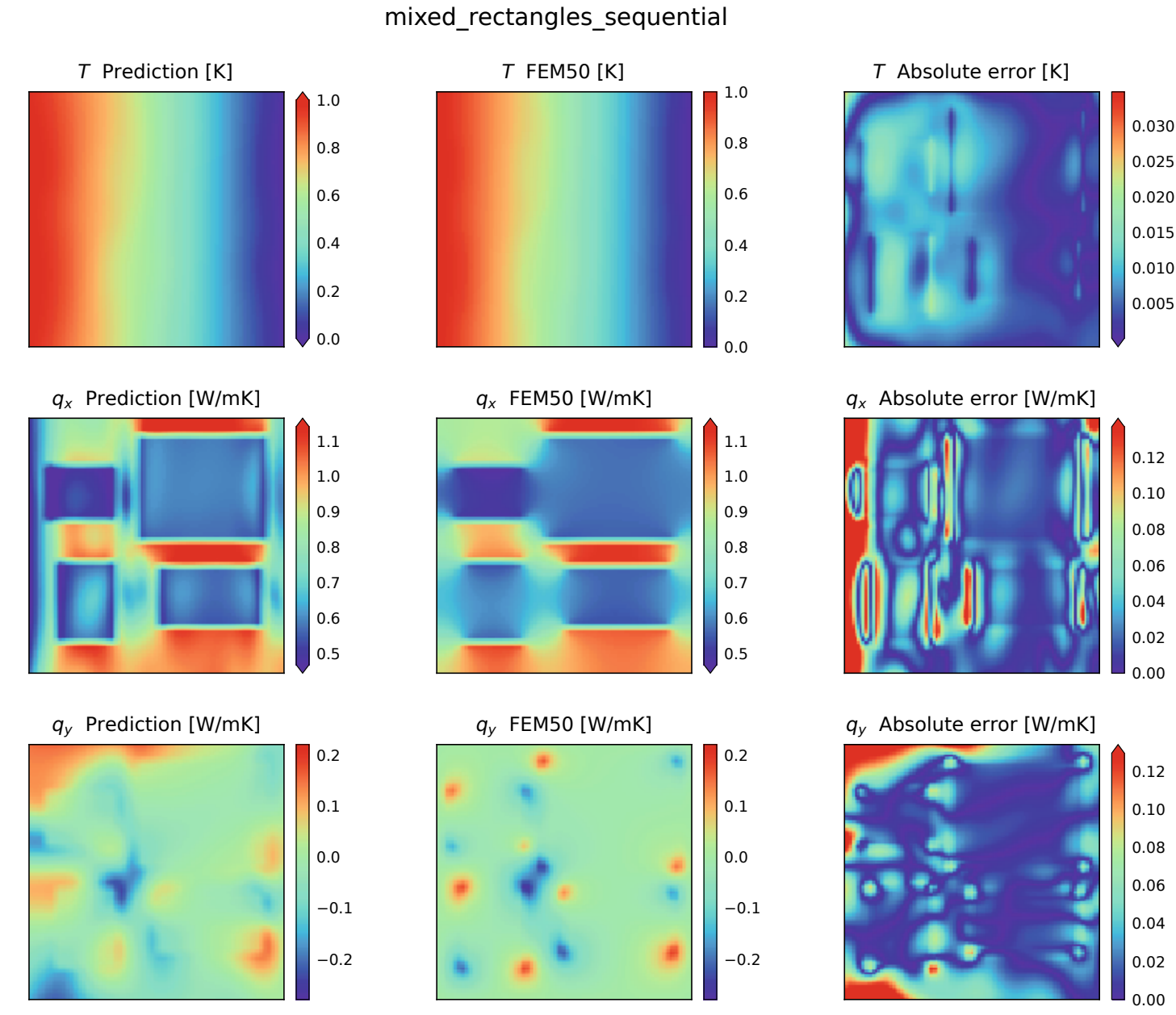




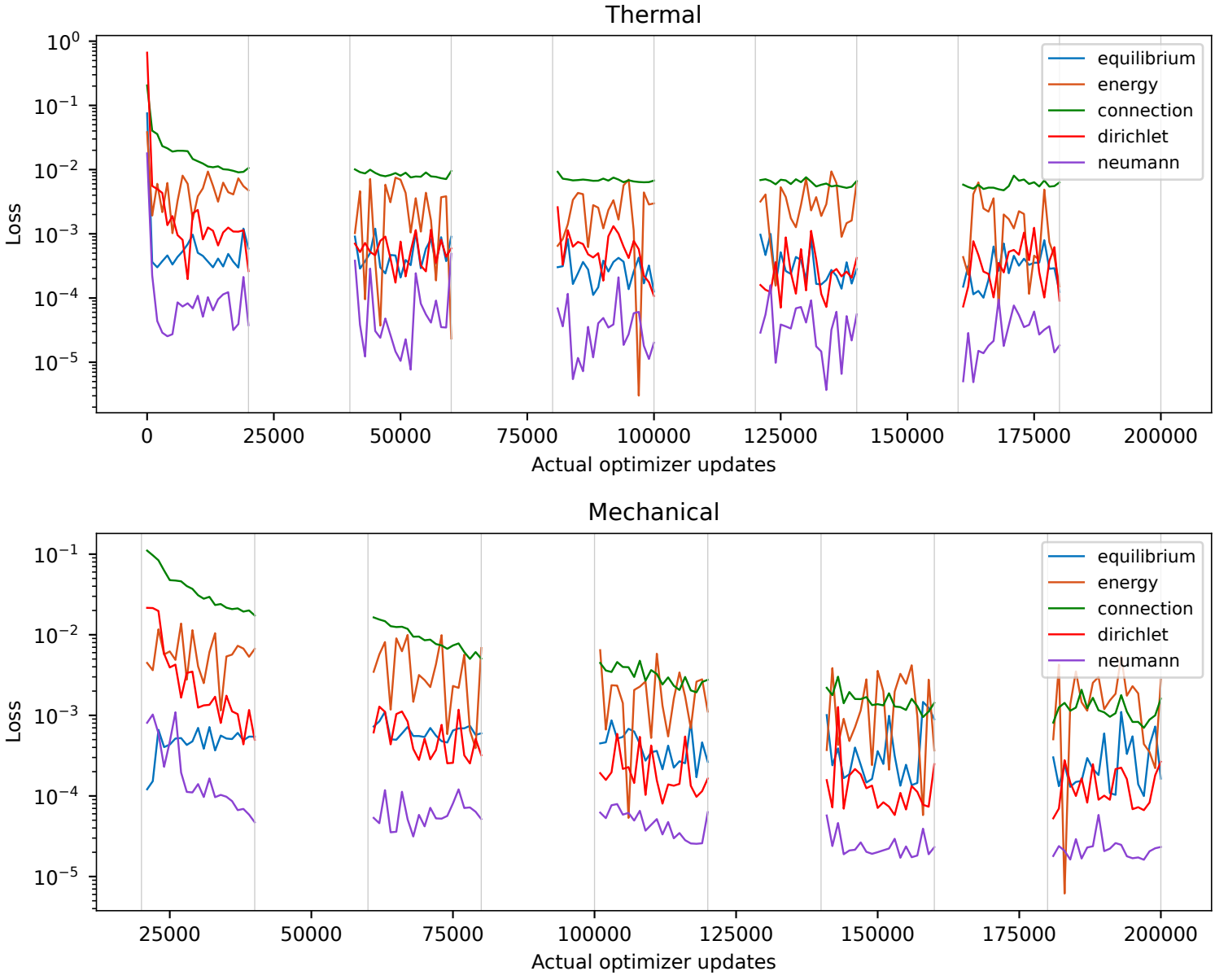
FULL CATALOGUE: verified completed jobs | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

mixed_rectangles_sequential

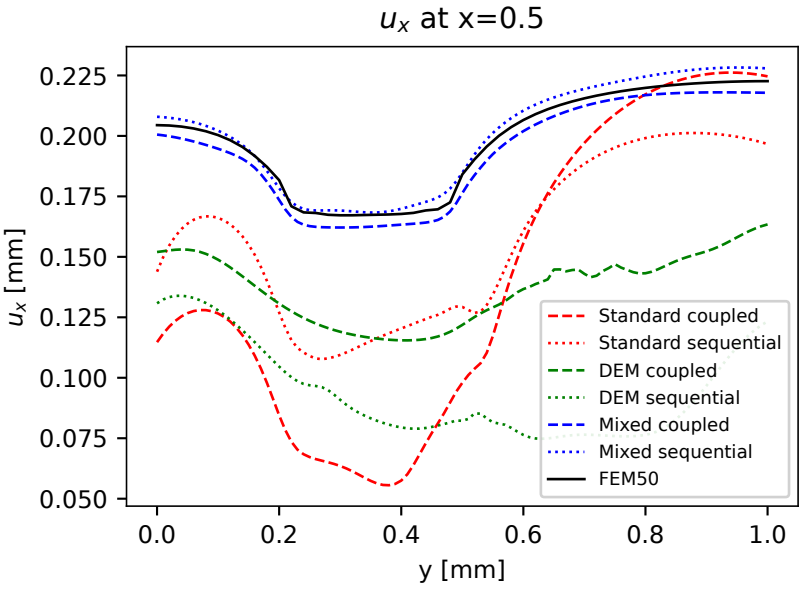


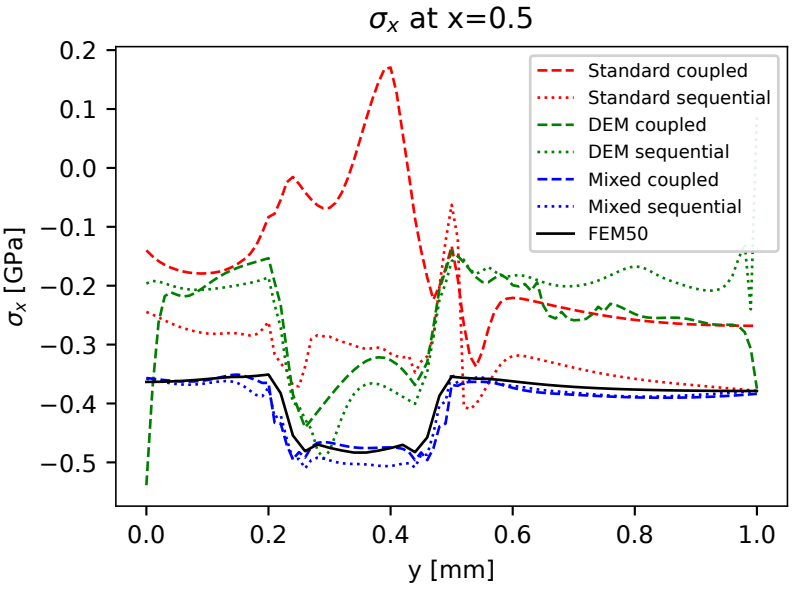


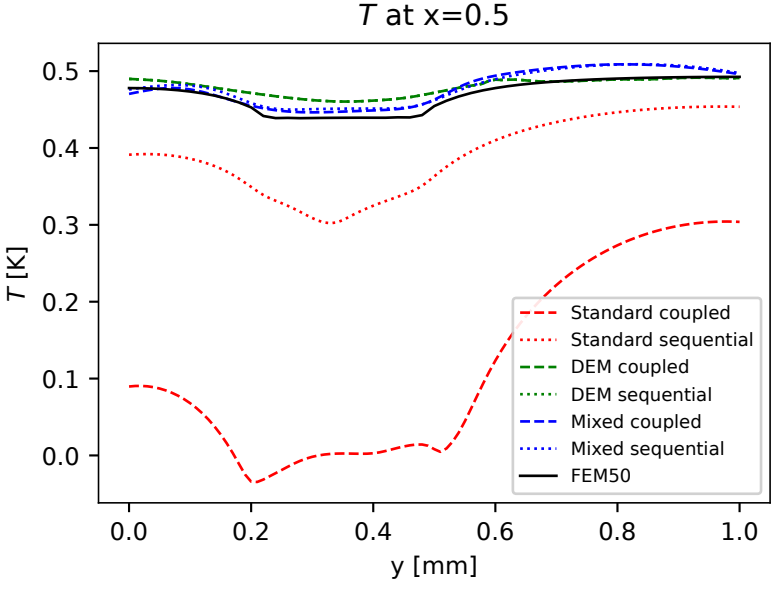
FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

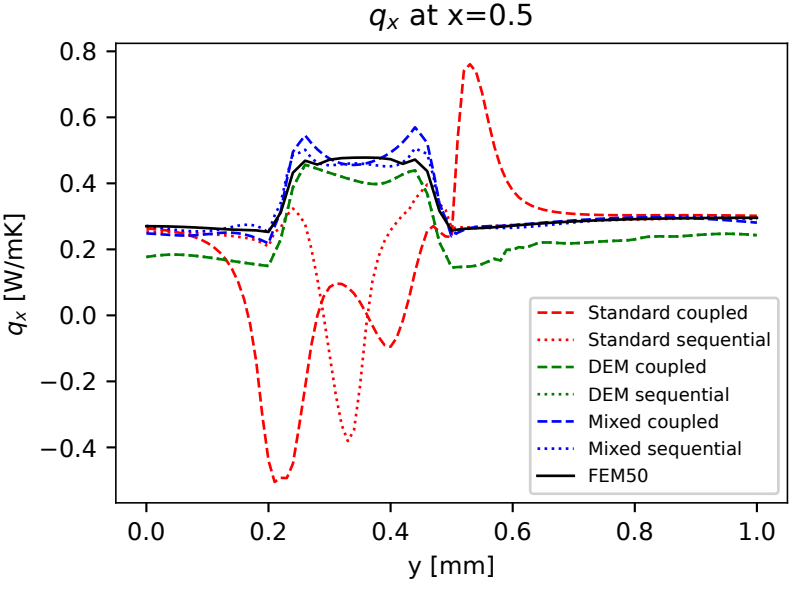


FULL CATALOGUE: verified completed jobs | Original logged samples only. Zero terms omitted on log axes; inactive terms are not filled or bridged.

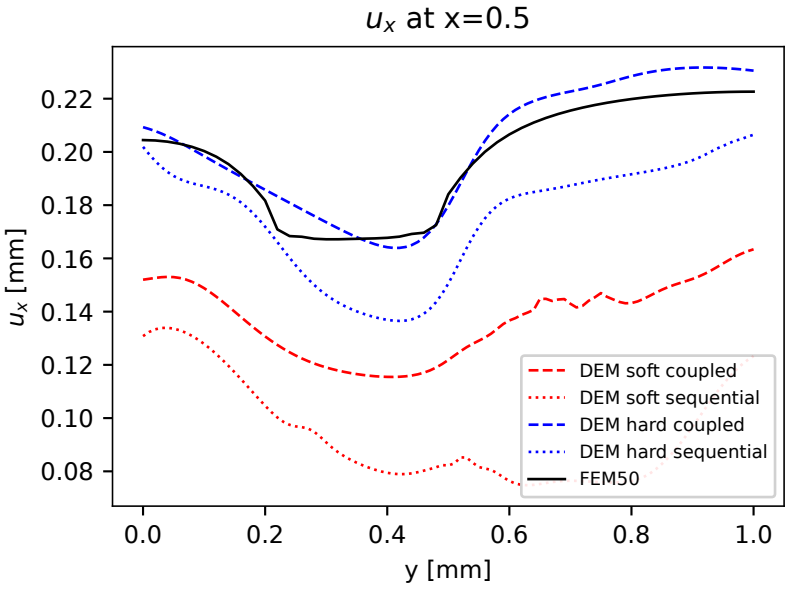


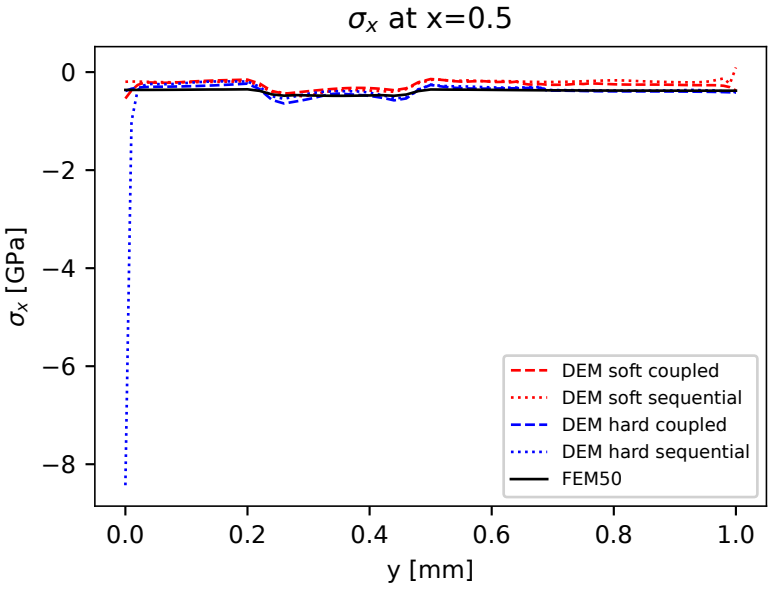


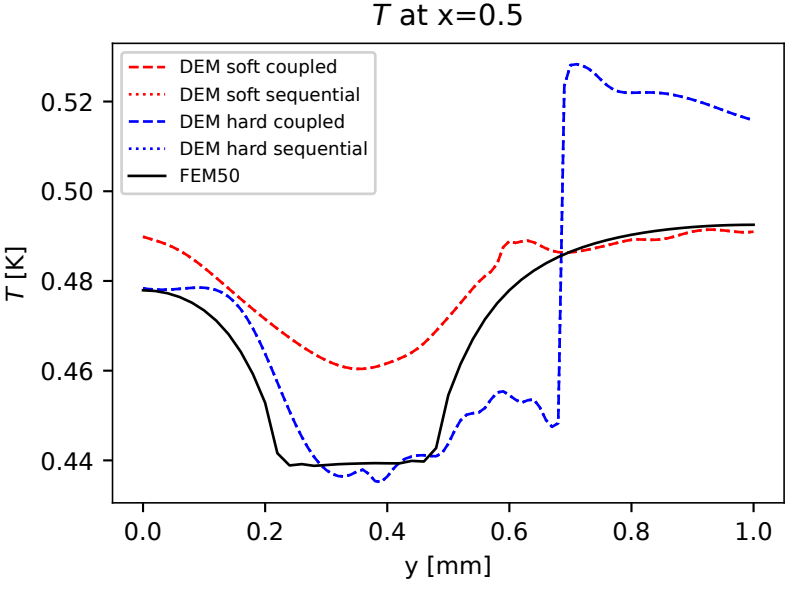


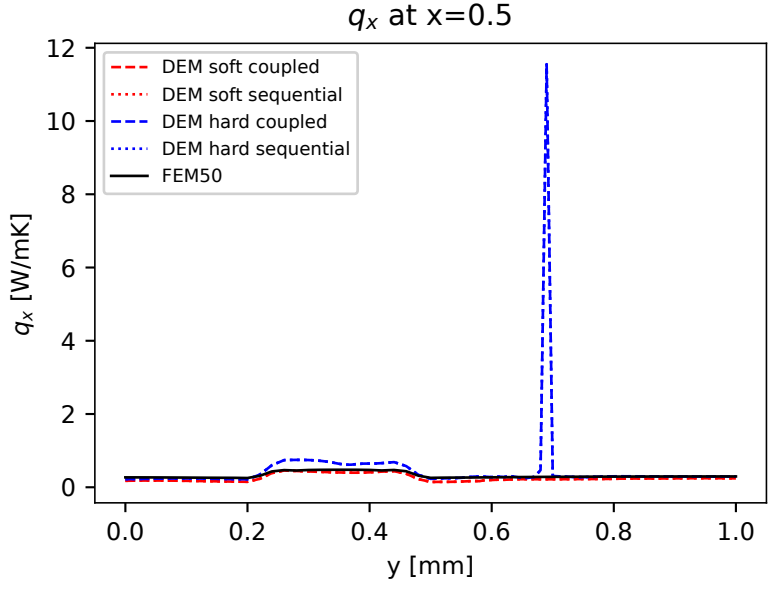


FULL CATALOGUE: verified completed jobs | All six completed full-budget cases, x=.5. Negative outcomes shown without ranking assumptions.

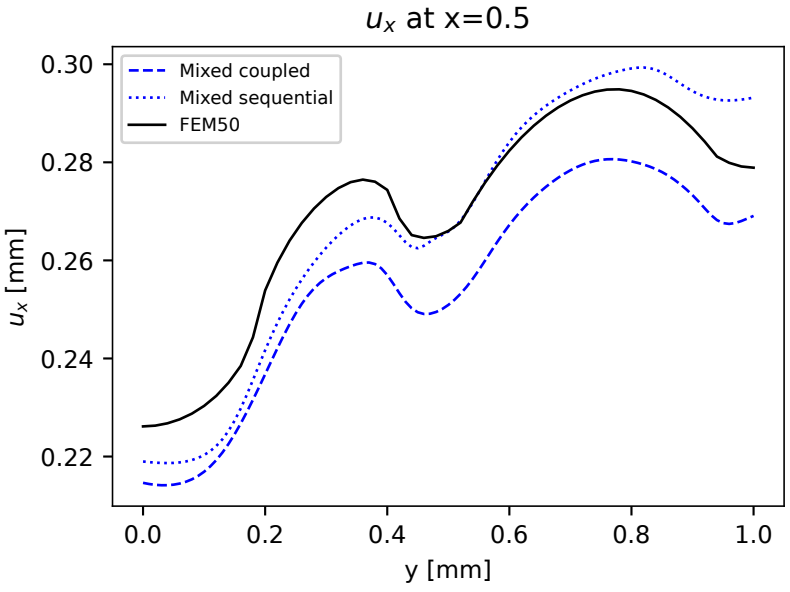


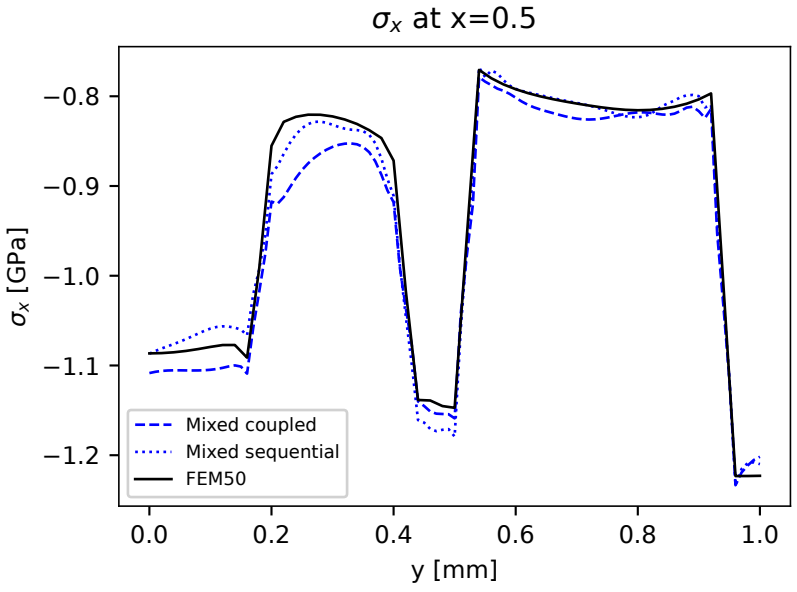


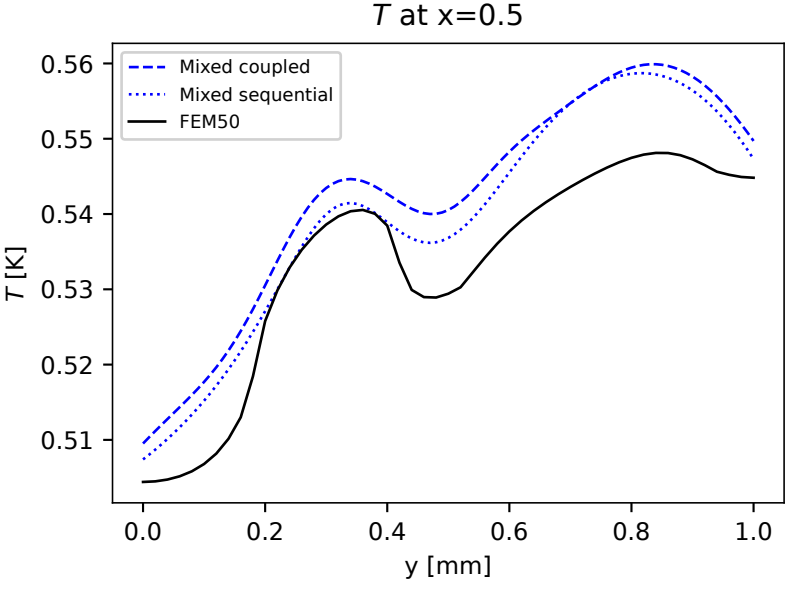


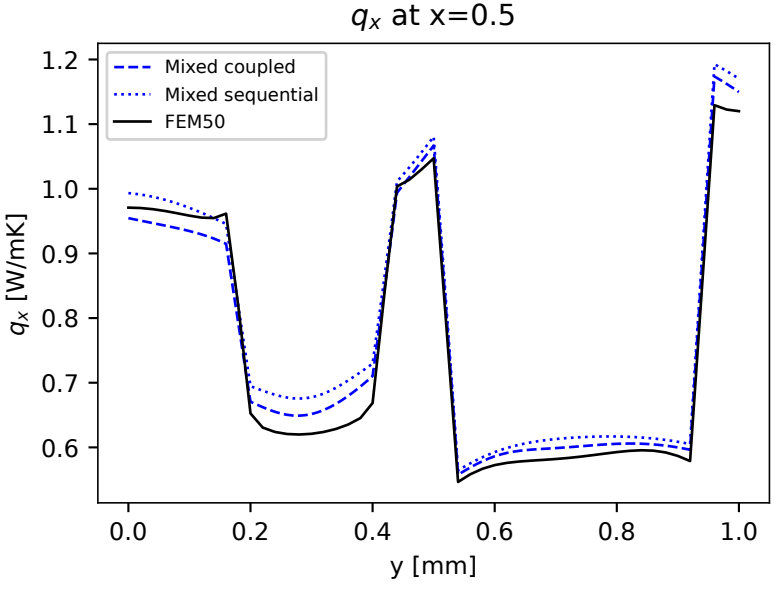


FULL CATALOGUE: verified completed jobs | Additional DEM soft/hard boundary sensitivity; not a separately reported manuscript experiment.

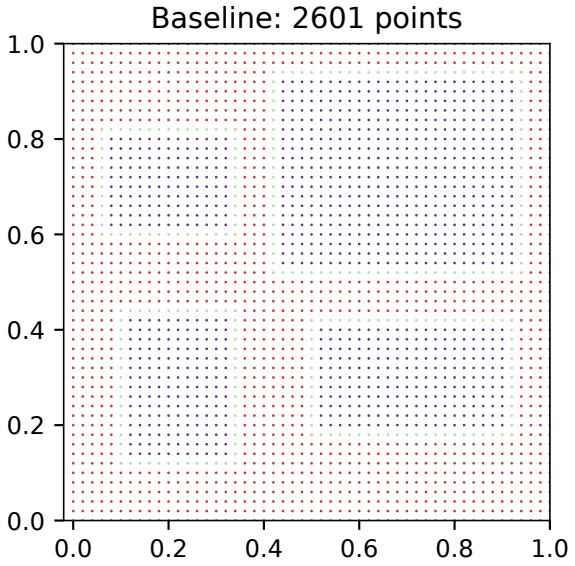


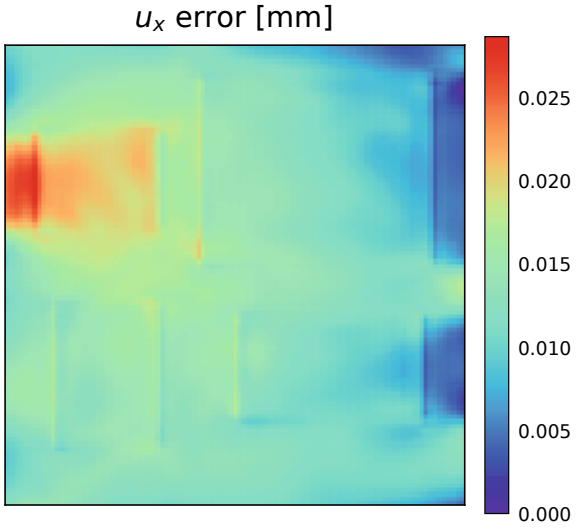


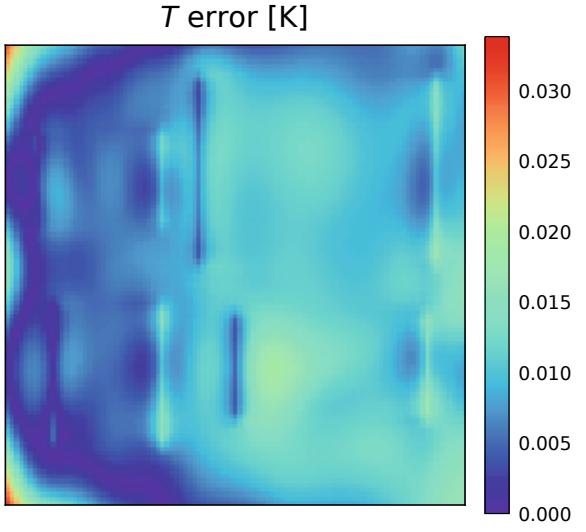


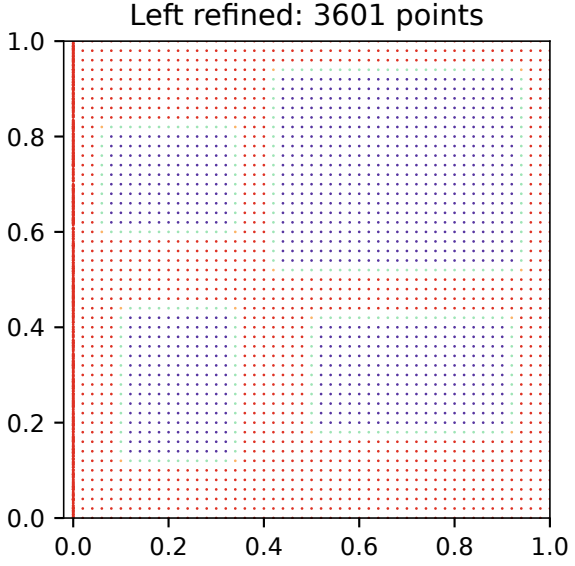


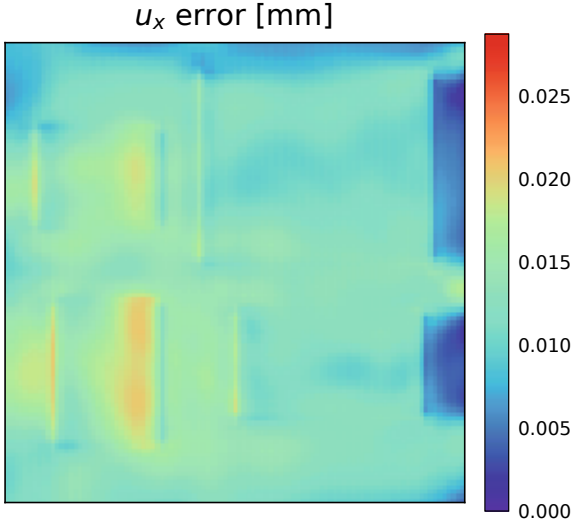
FULL CATALOGUE: verified completed jobs | Saved full-budget coupled and sequential predictions; no fitted scaling.

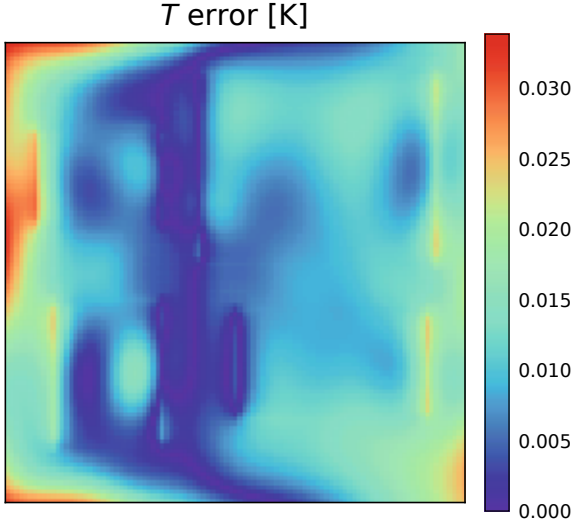




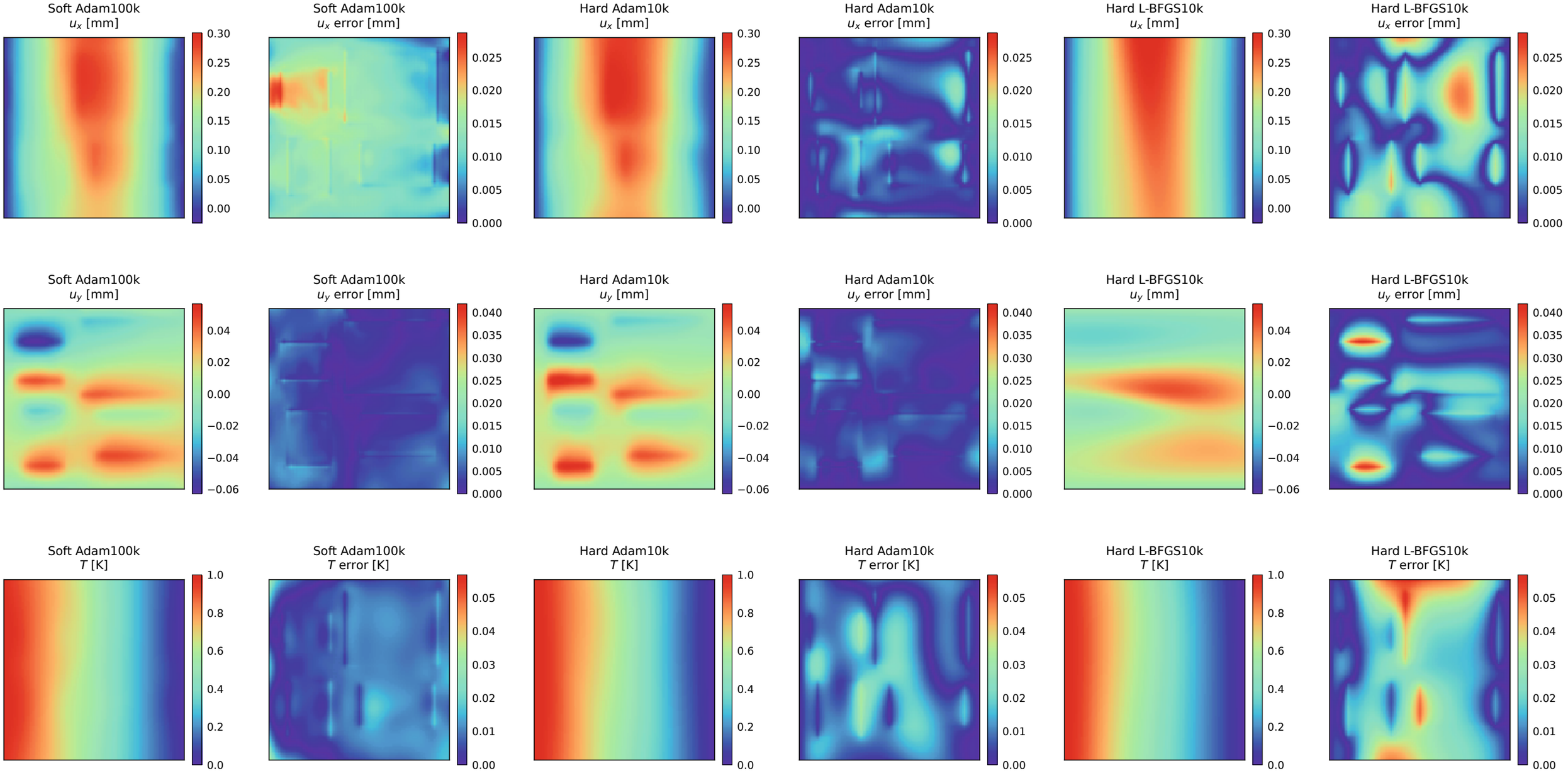




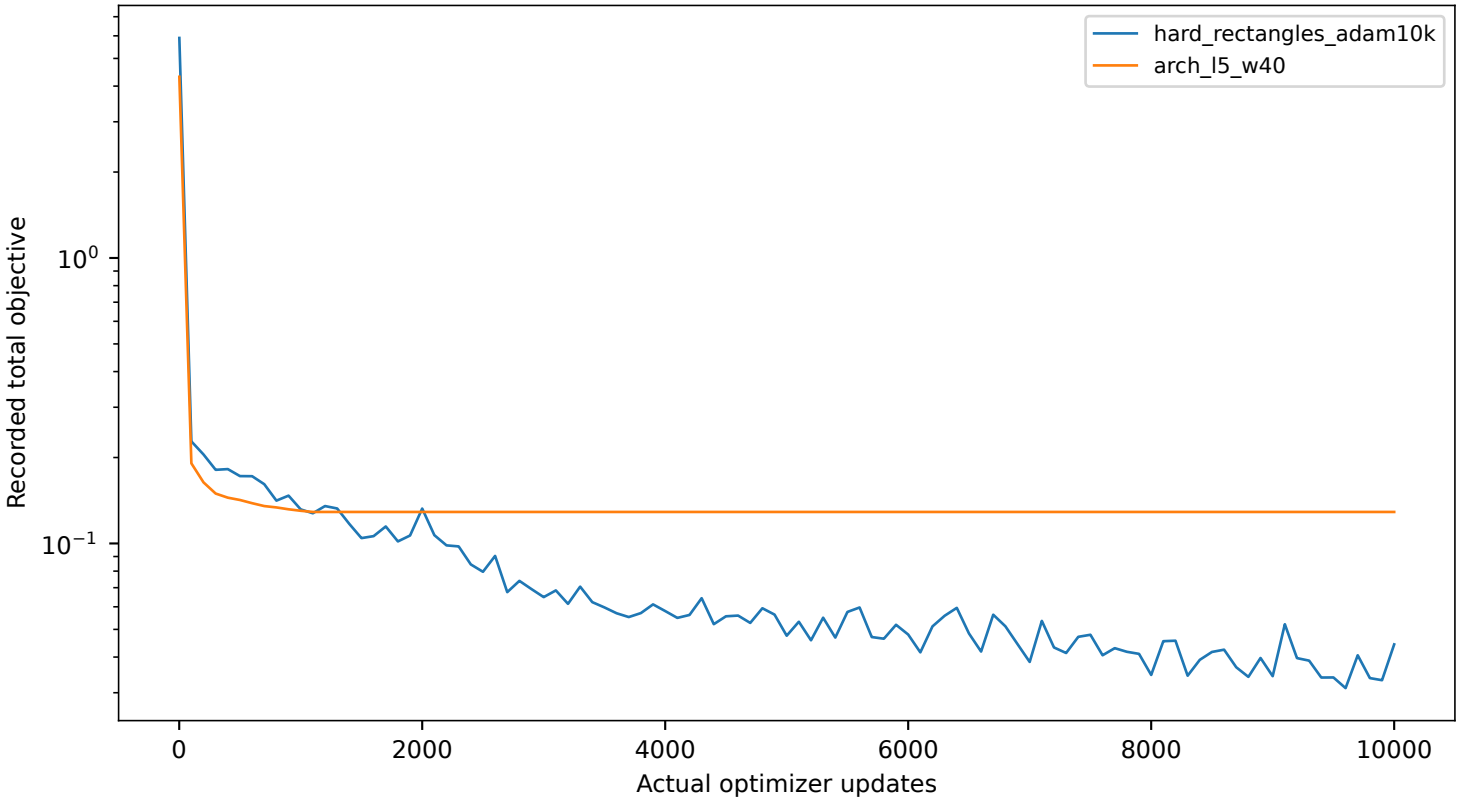




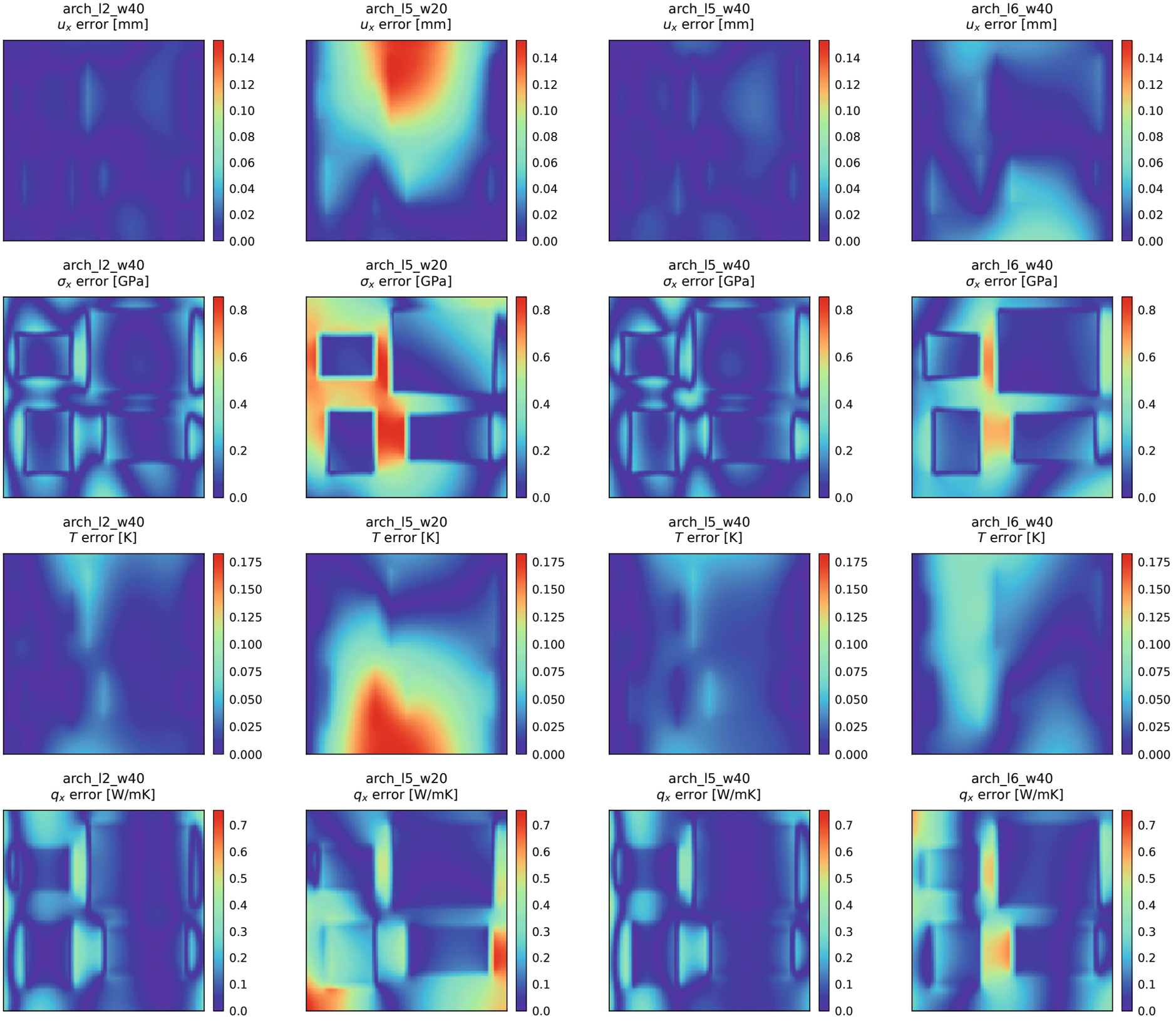
FULL CATALOGUE: verified completed jobs | Same deterministic sampler;1000left-edge points added. Shared error scales; actual outcomes may improve or deteriorate.



FULL CATALOGUE: verified completed jobs | Field/error pairs. Soft Adam100k versus hard Adam10k and hard L-BFGS10k; unequal budgets disclosed.

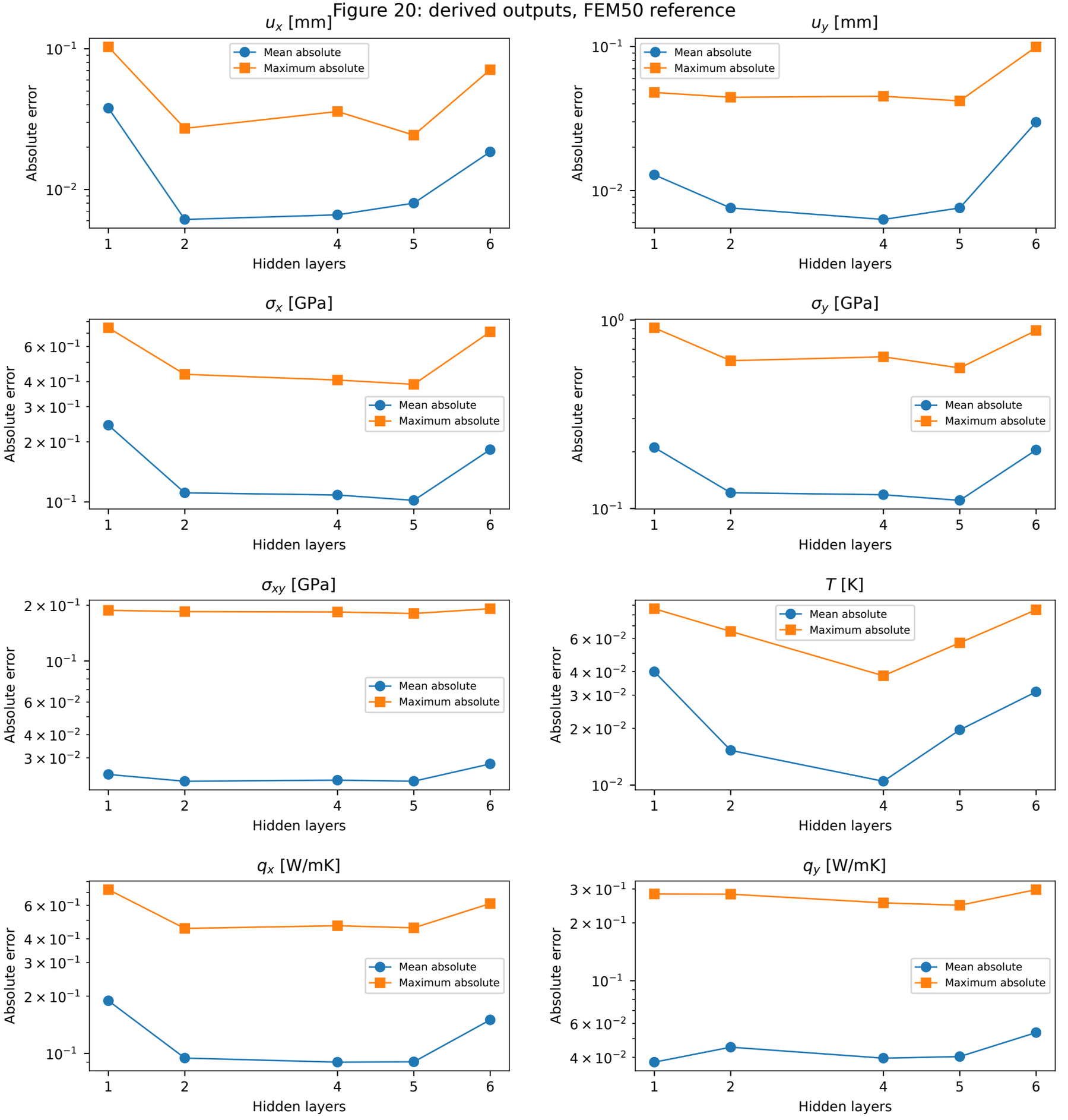


FULL CATALOGUE: verified completed jobs | Original logged objectives; Adam and L-BFGS logging evaluation moments may differ. No resampling.



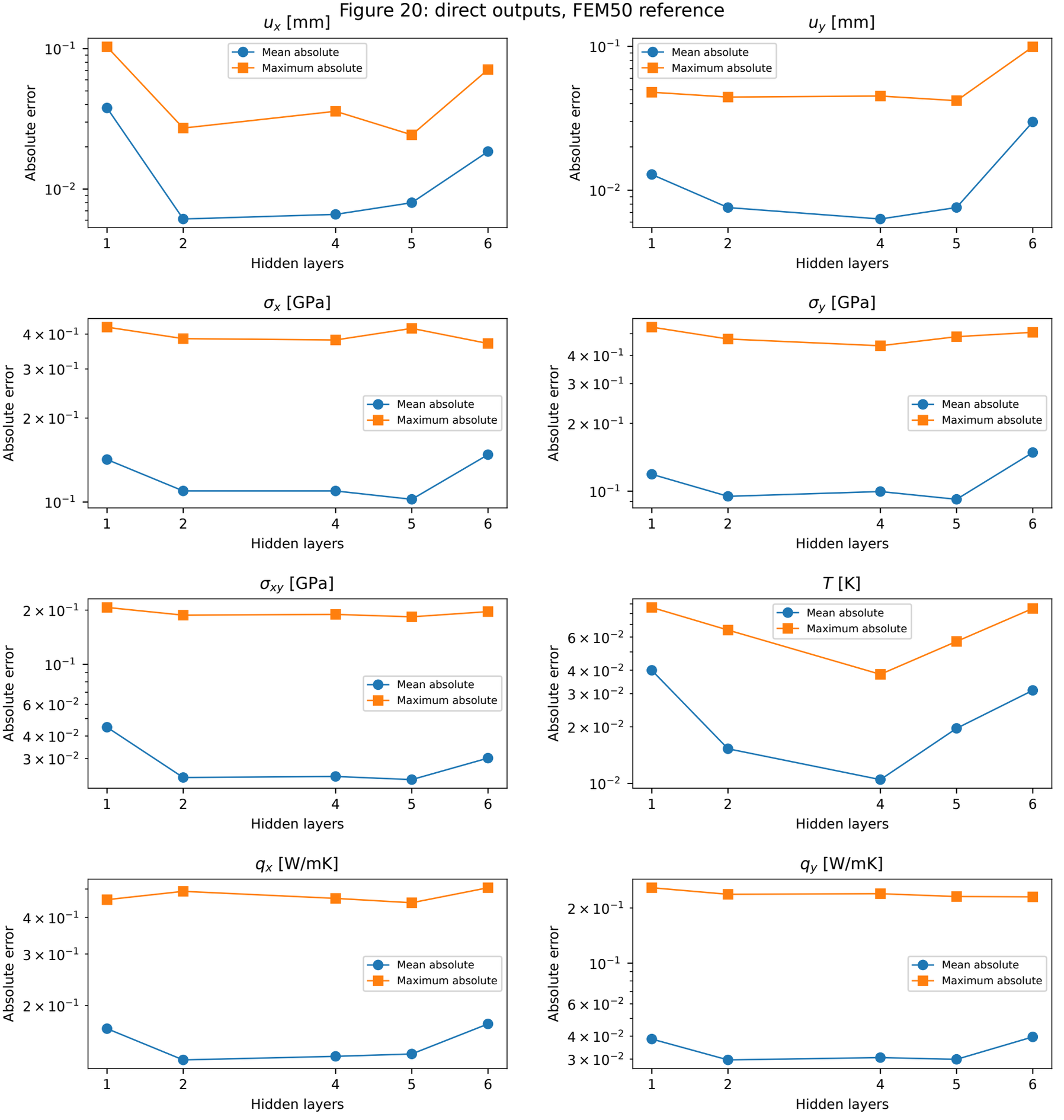
FULL CATALOGUE: verified completed jobs | Selected2x40,5x20,5x40,6x40; each10k L-BFGS. Shared per-field limits across cases; derived fields.

Figure 20: derived outputs, FEM50 reference



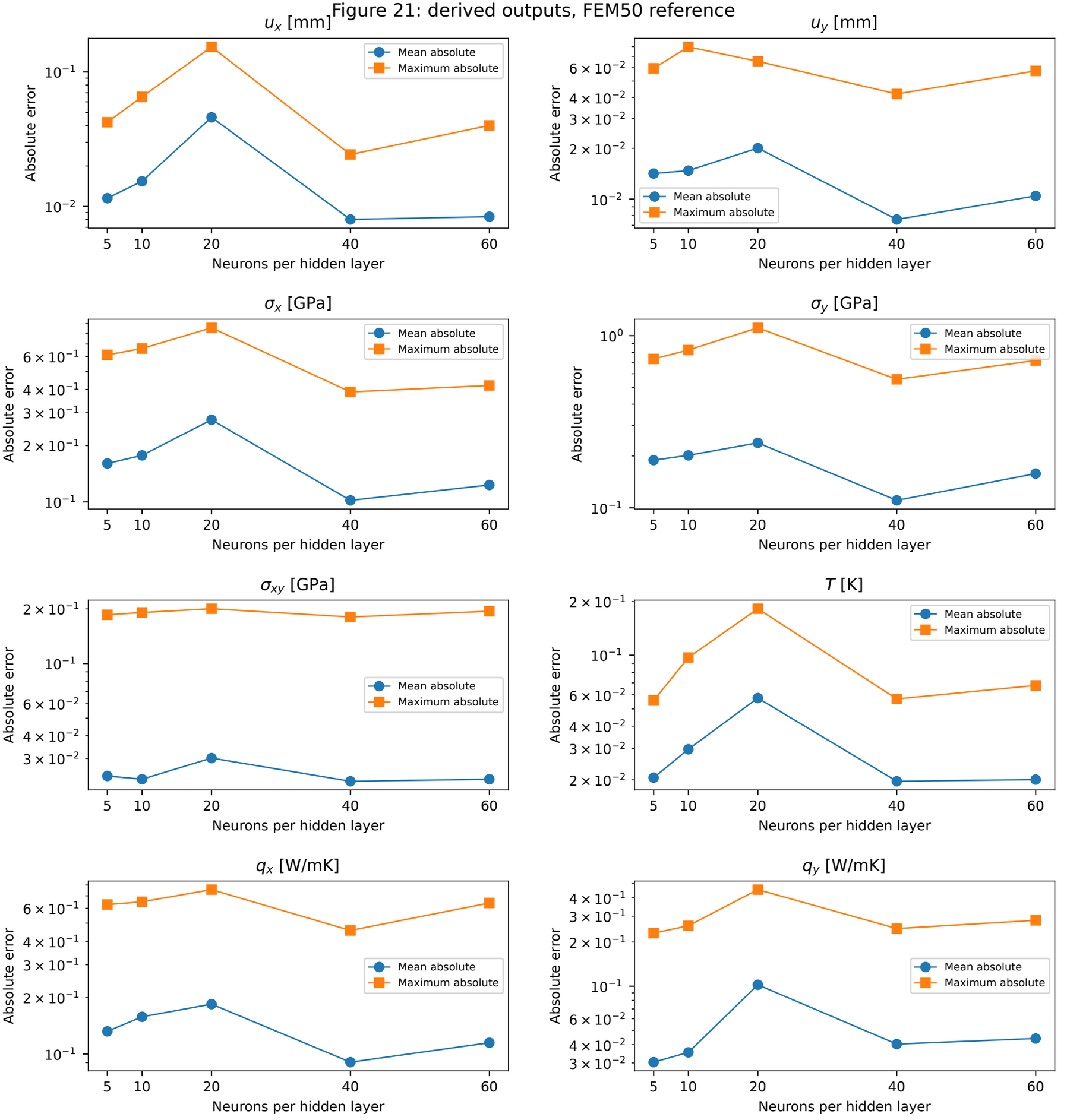
FULL CATALOGUE: verified completed jobs | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Figure 20: direct outputs, FEM50 reference



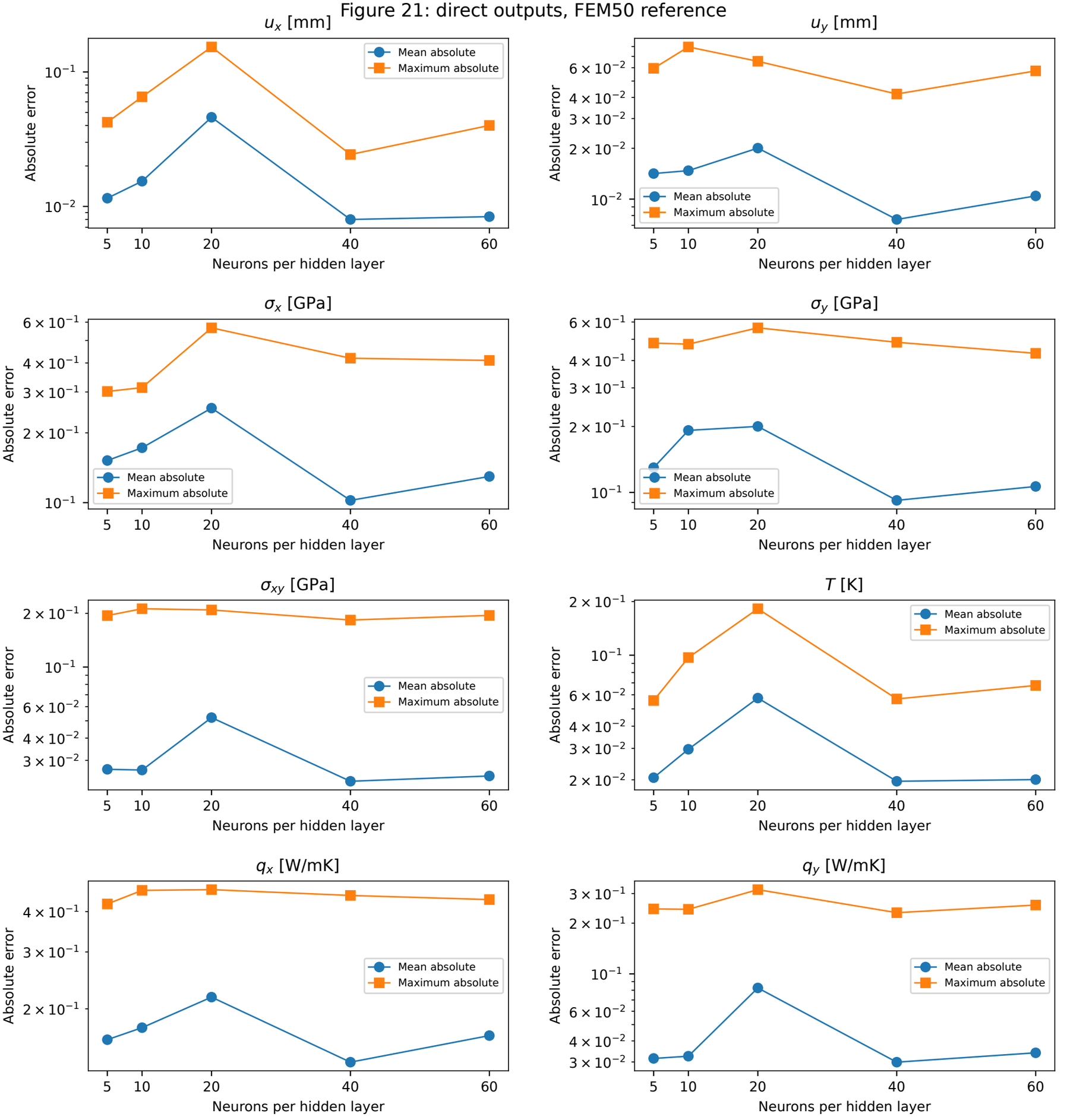
FULL CATALOGUE: verified completed jobs | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Figure 21: derived outputs, FEM50 reference



FULL CATALOGUE: verified completed jobs | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Figure 21: direct outputs, FEM50 reference



FULL CATALOGUE: verified completed jobs | All eight fields. Maximum and mean measured errors; no assumption of monotonic improvement.

Table 4 reproduction: seconds / iteration through first100 L-BFGS updates
Rows: hidden layers; columns: neurons per layer

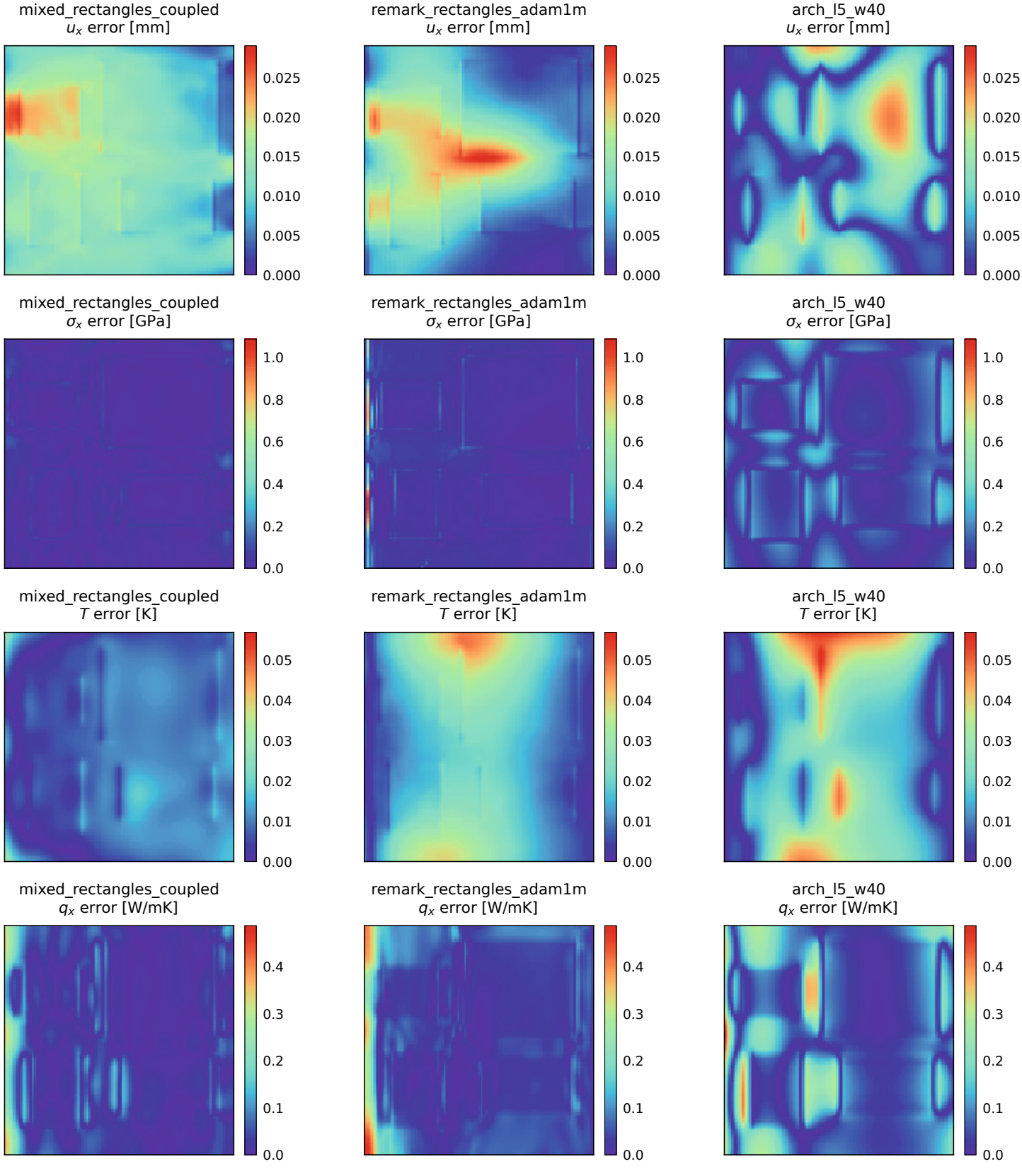
	5	10	20	40	60
1	0.0634942	0.0626547	0.0681982	0.296767	0.25935
2	0.307545	0.0805924	0.0907312	0.0804002	0.0902531
4	0.102286	0.0893592	0.116194	0.0975289	0.0987992
5	0.293395	0.34495	0.46897	0.120403	0.113849
6	0.52045	0.119539	0.541844	0.621843	0.527139

FULL CATALOGUE: verified completed jobs | Wall time, not isolated GPU time. cuda / NVIDIA GeForce RTX 3090.

Remark7: separately measured100k /1M Adam and10k hard L-BFGS

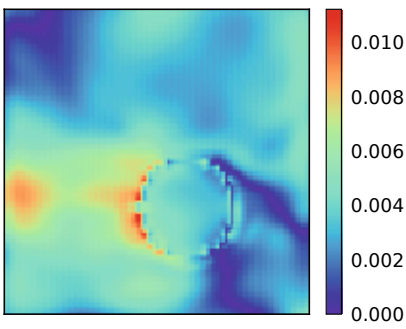
Case	Updates	Wall seconds	Device
mixed_rectangles_coupled	100,000	2780	NVIDIA GeForce RTX 3090
remark_rectangles_adam1m	1,000,000	2.539e+04	NVIDIA GeForce RTX 3090
arch_l5_w40	10,000	3586	NVIDIA GeForce RTX 3090

FULL CATALOGUE: verified completed jobs | 1M continues the100k checkpoint; original100k remains unchanged. Different hardware from paper; wall-time overhead included.

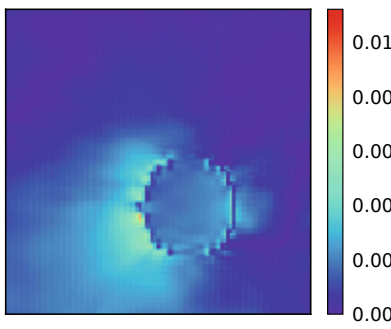


FULL CATALOGUE: verified completed jobs | Separate1M continuation versus100k baseline and10k hard L-BFGS; main baseline figures still100k. Shared per-field limits across cases; derived fields.

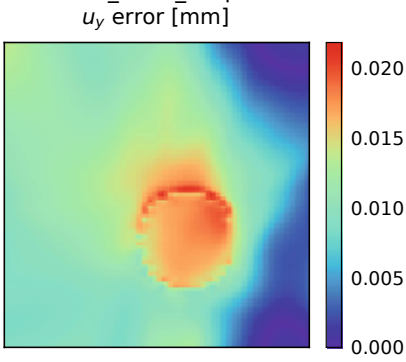
mixed_circle_coupled
u_x error [mm]



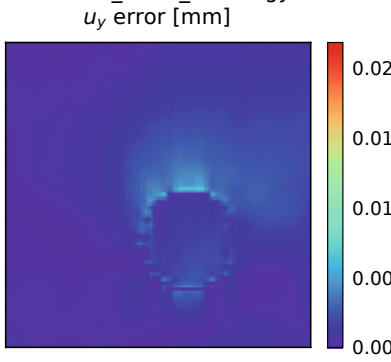
ablation_circle_noenergy
u_x error [mm]



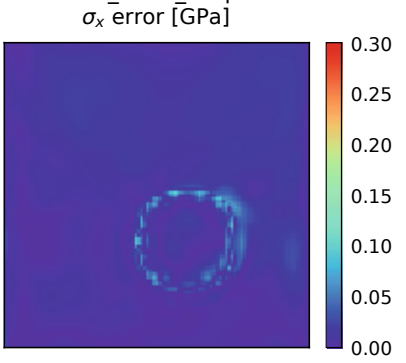
mixed_circle_coupled
u_y error [mm]



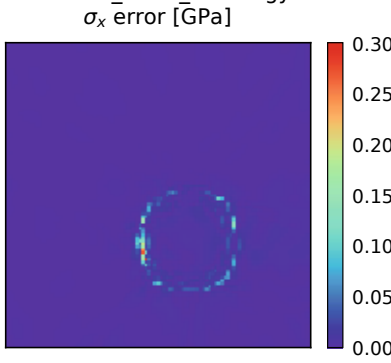
ablation_circle_noenergy
u_y error [mm]



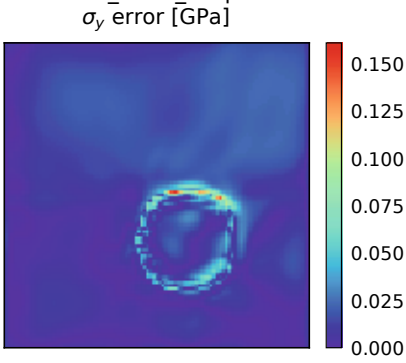
mixed_circle_coupled
σ_x error [GPa]



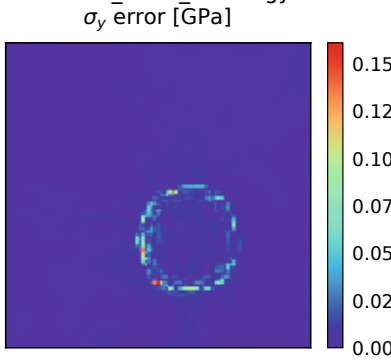
ablation_circle_noenergy
σ_x error [GPa]



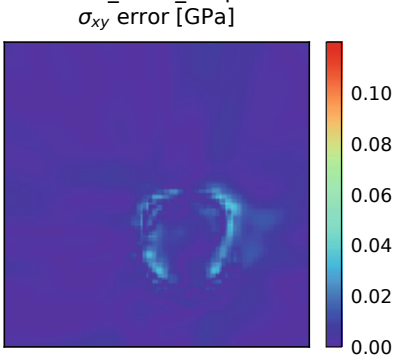
mixed_circle_coupled
σ_y error [GPa]



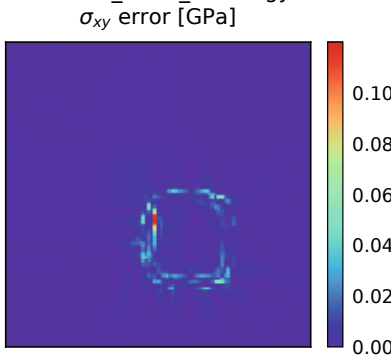
ablation_circle_noenergy
σ_y error [GPa]



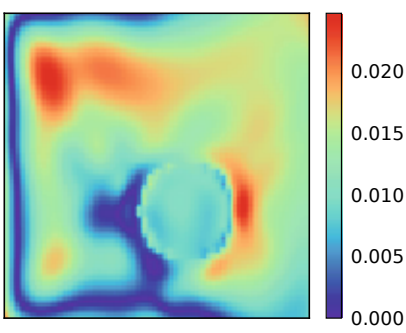
mixed_circle_coupled
σ_{xy} error [GPa]



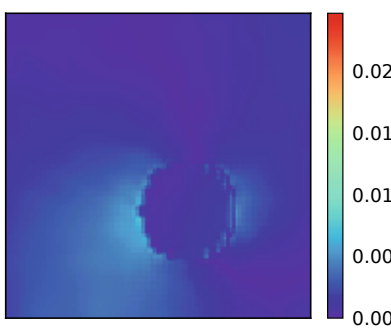
ablation_circle_noenergy
σ_{xy} error [GPa]



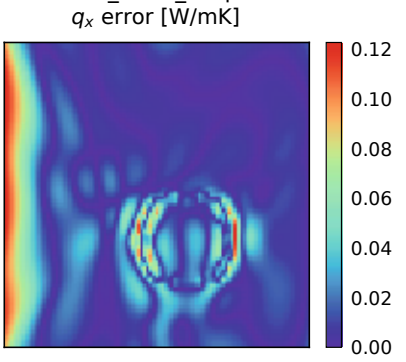
mixed_circle_coupled
T error [K]



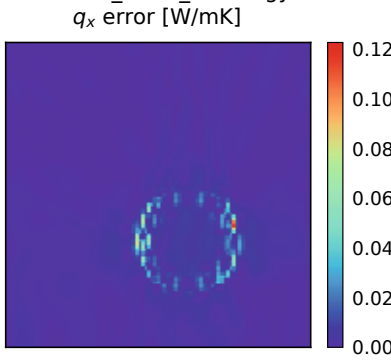
ablation_circle_noenergy
T error [K]



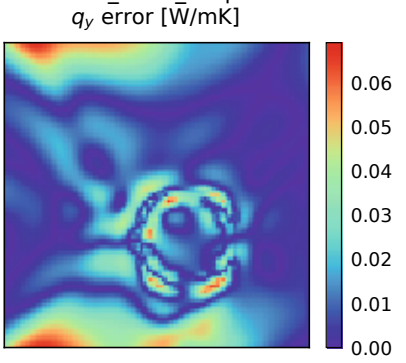
mixed_circle_coupled
q_x error [W/mK]



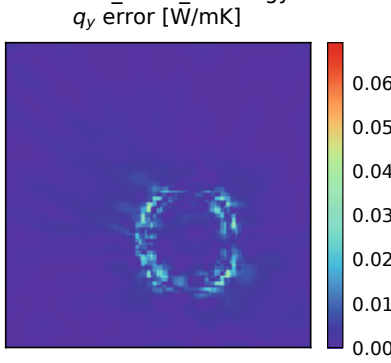
ablation_circle_noenergy
q_x error [W/mK]



mixed_circle_coupled
q_y error [W/mK]

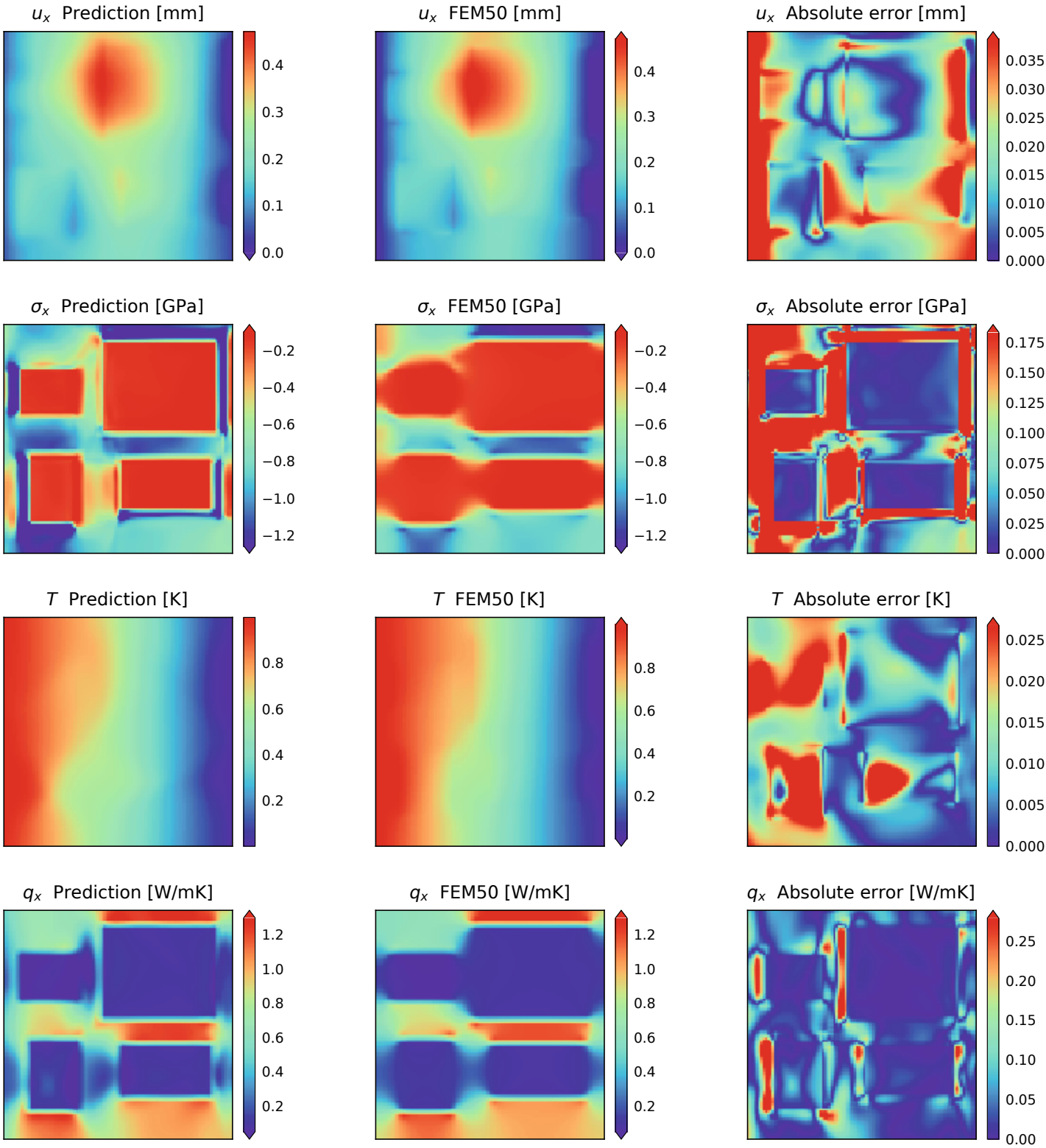


ablation_circle_noenergy
q_y error [W/mK]



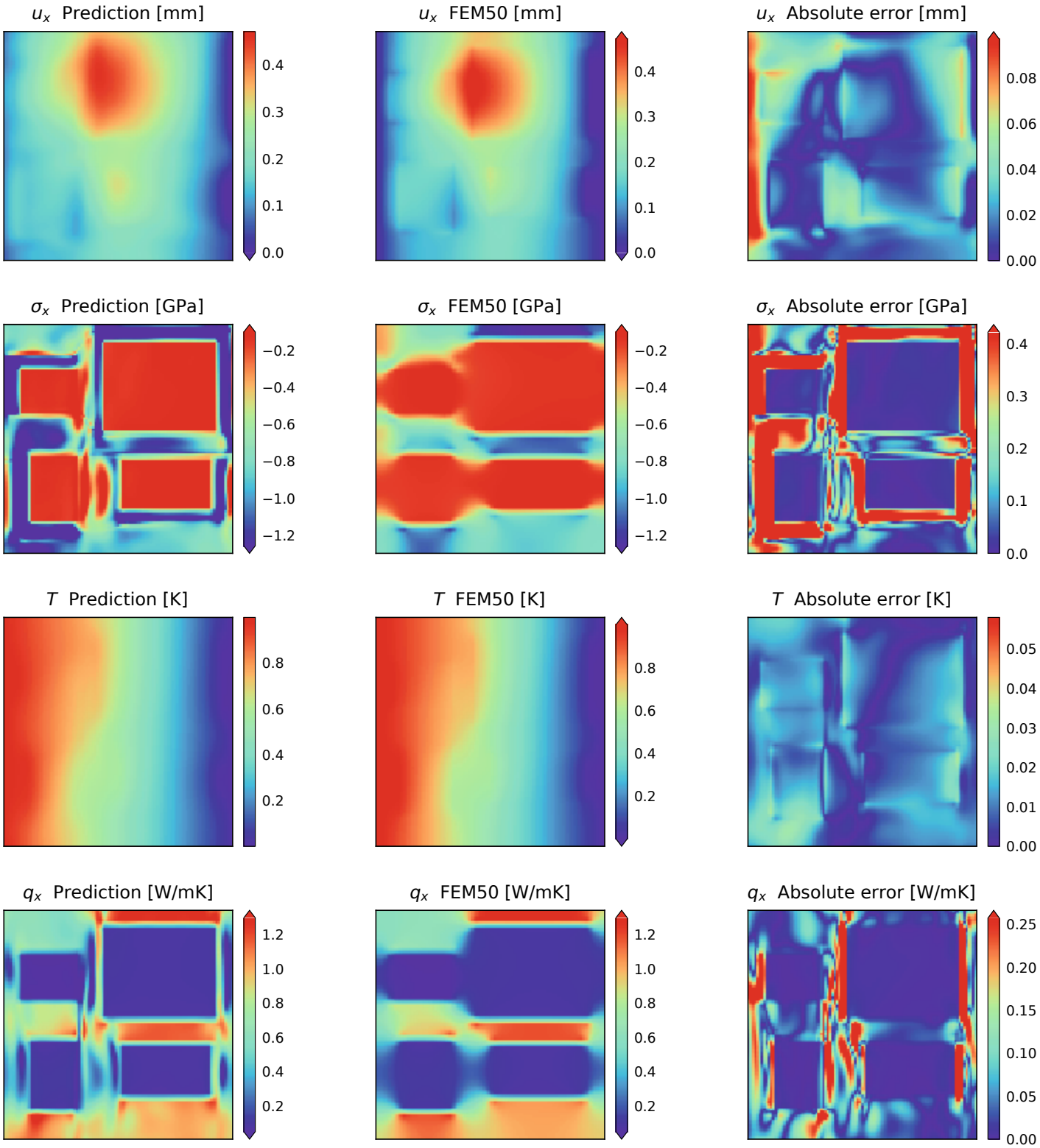
FULL CATALOGUE: verified completed jobs | Energy omission is a reconstruction of the unquantified paper claim. Shared per-field limits across cases; derived fields.

Held-out ratio15: transfer_w0.1

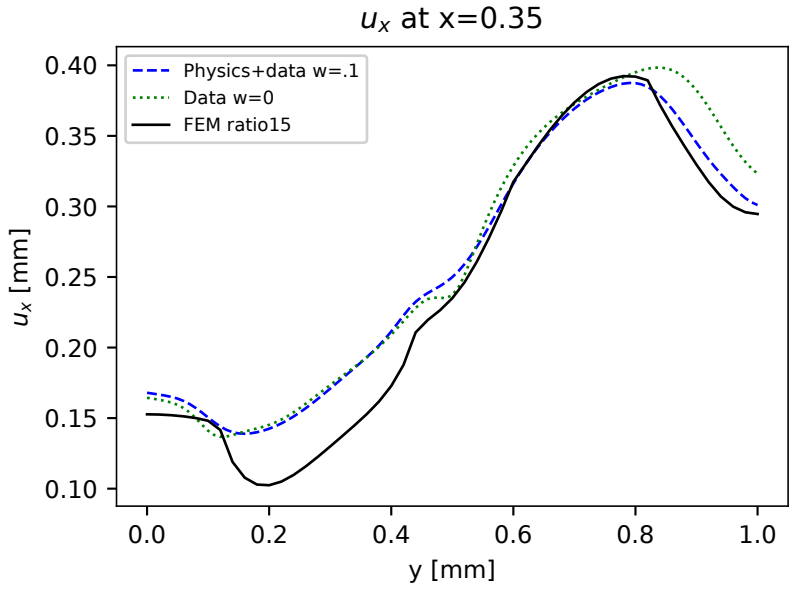


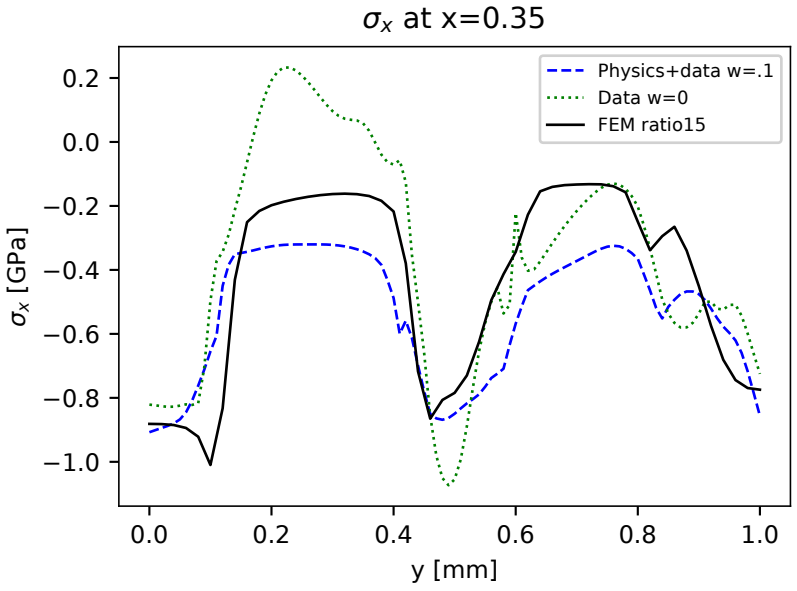
FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

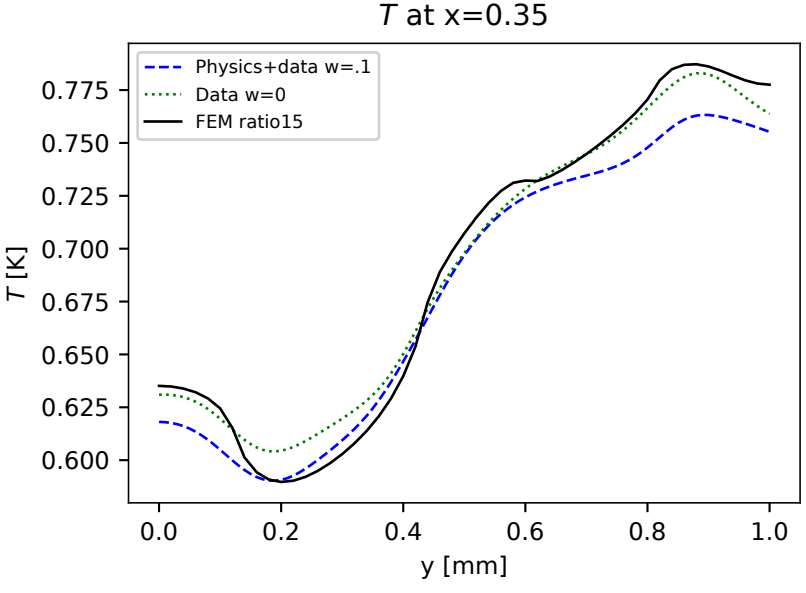
Held-out ratio15: transfer_w0

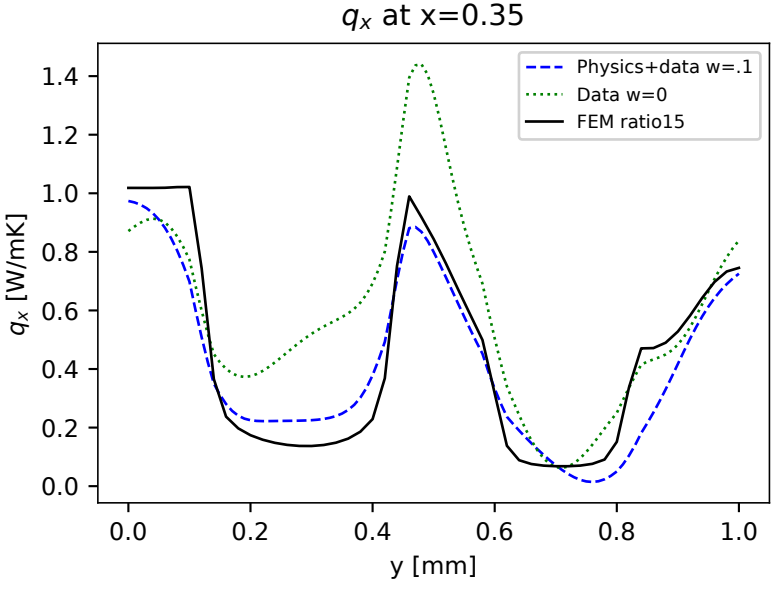


FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

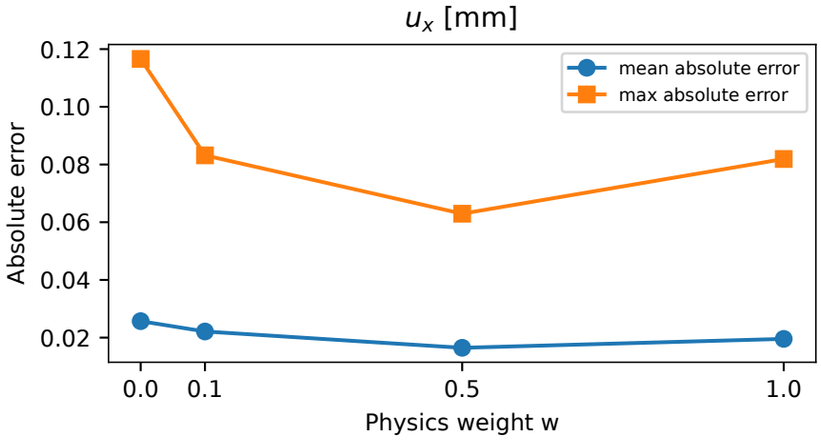


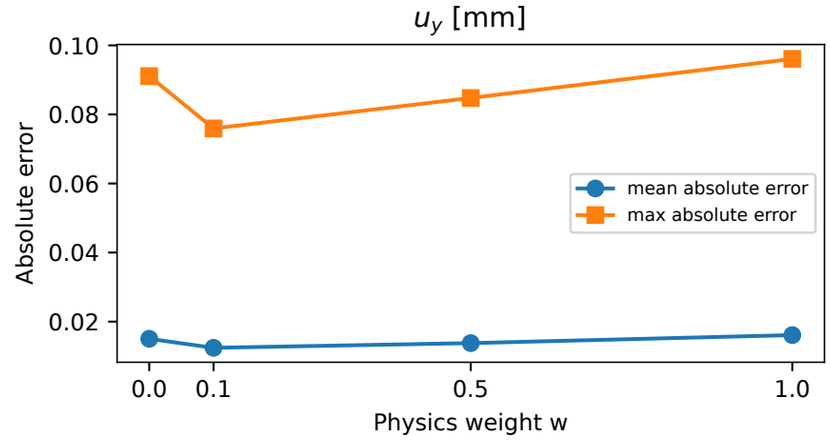


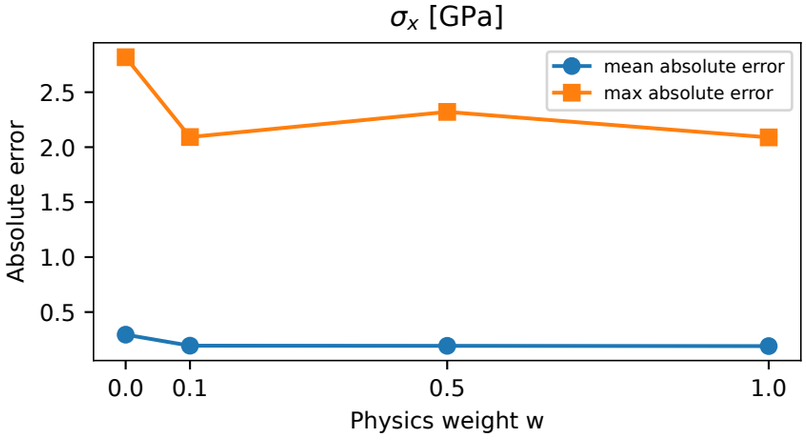


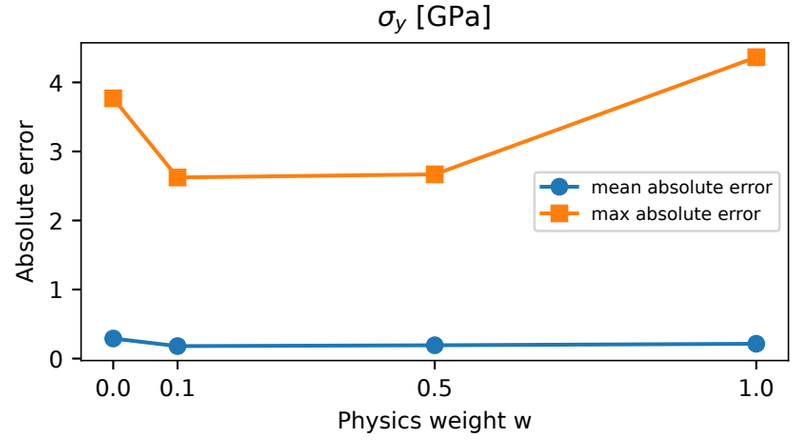


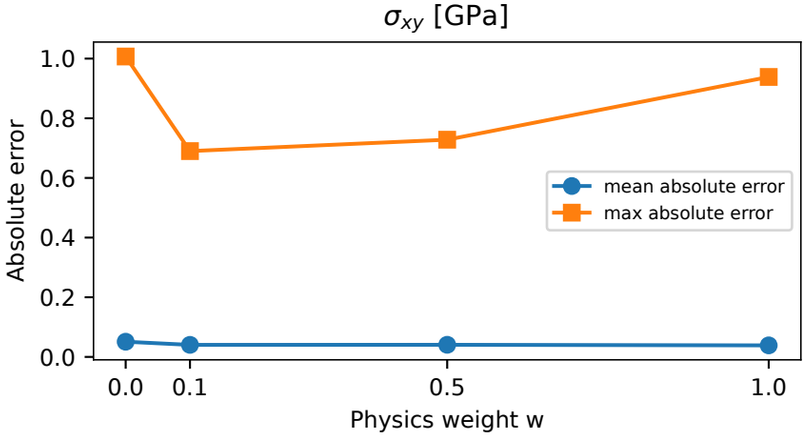
FULL CATALOGUE: verified completed jobs | x=.35 follows panel titles and body; paper caption says.5. Held-out ratio15, no retraining.

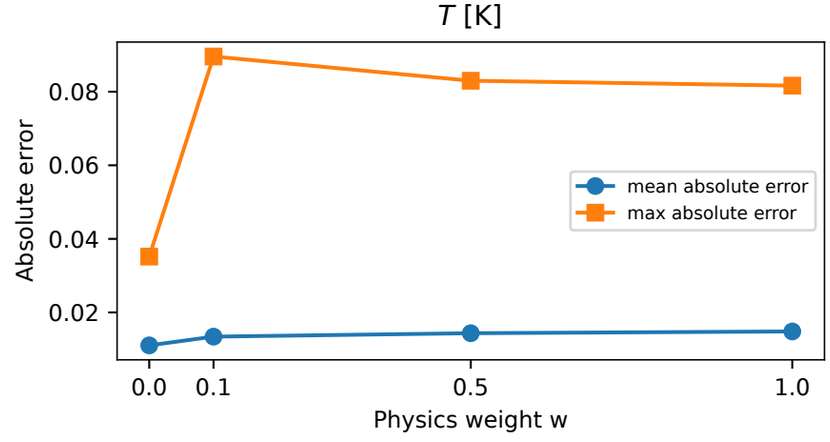


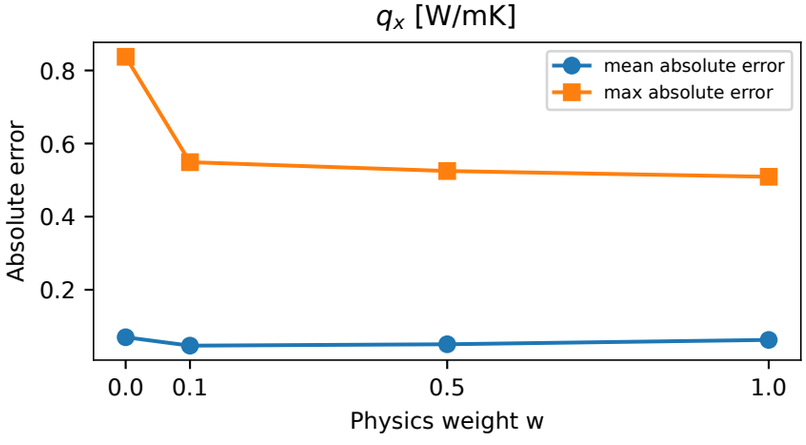


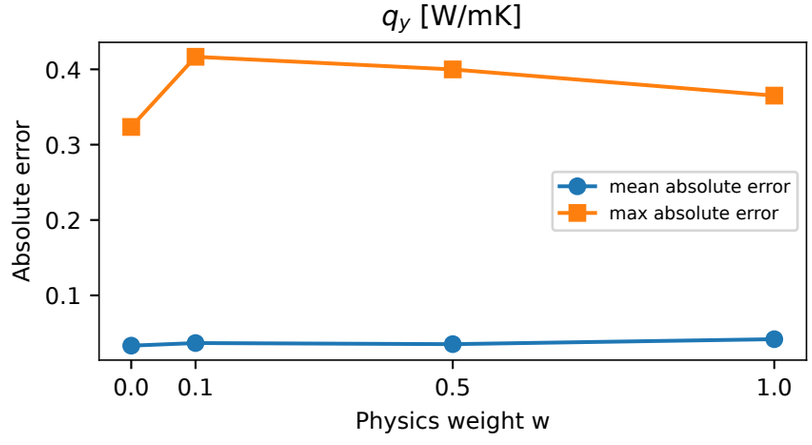






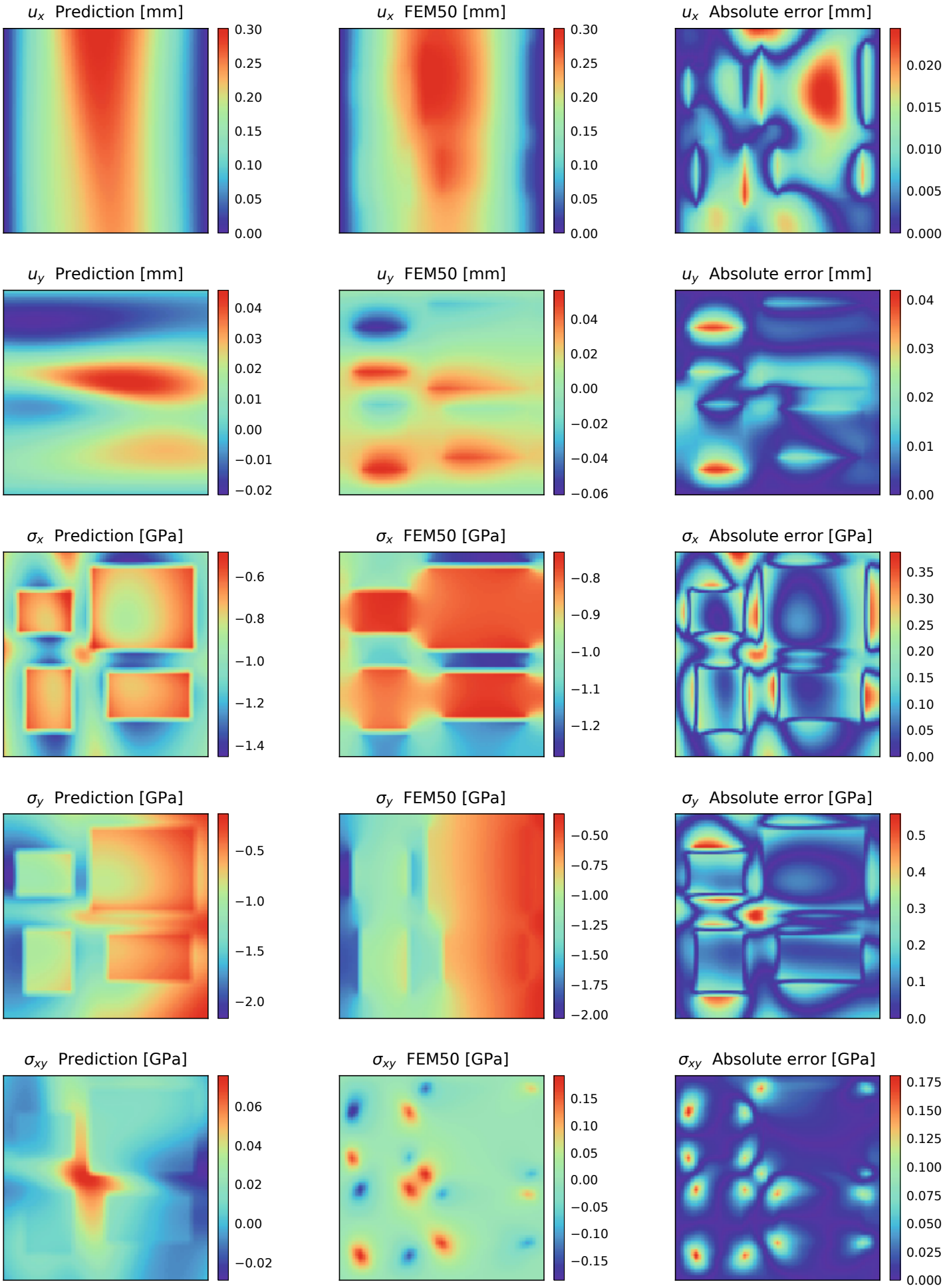




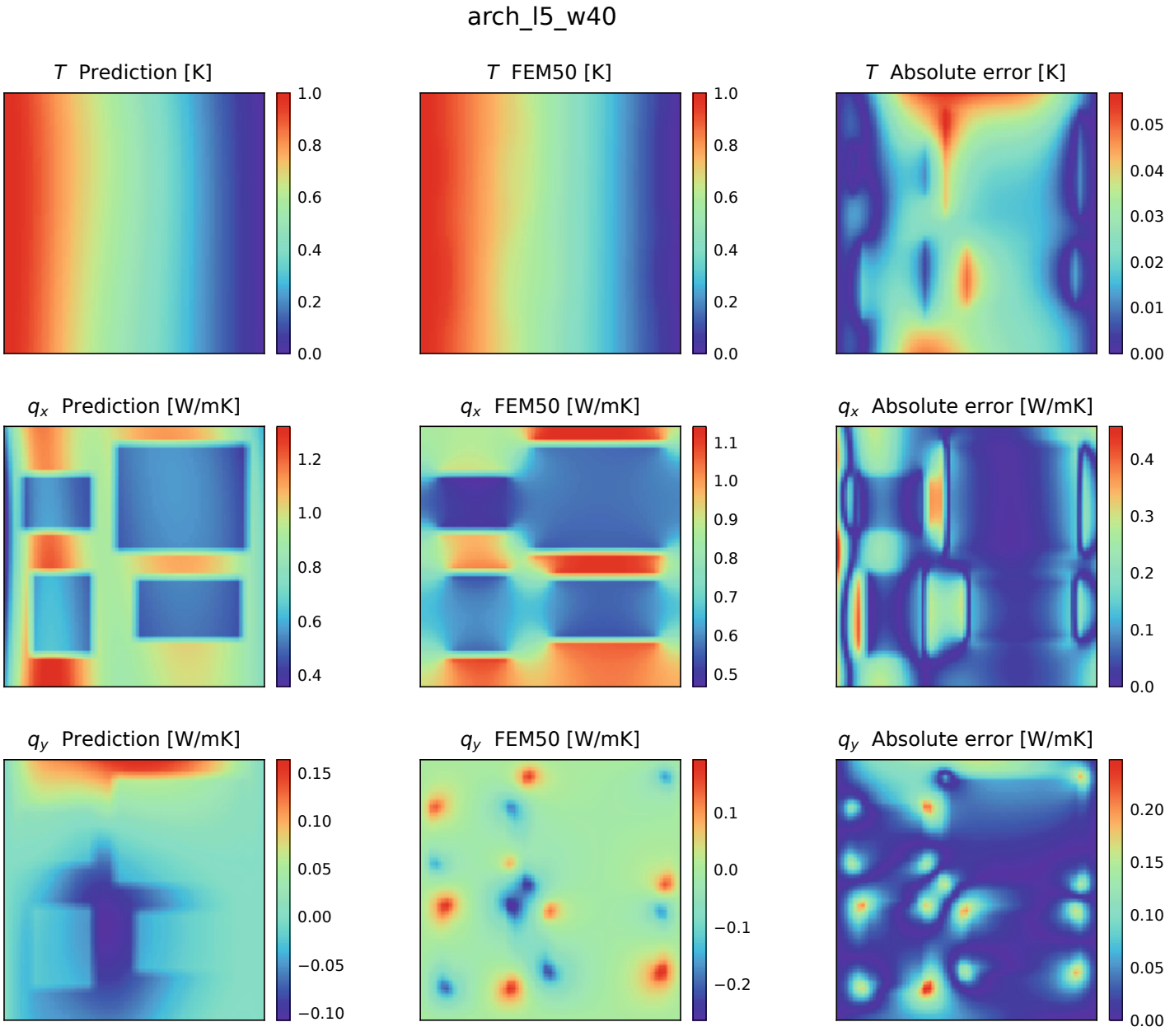


FULL CATALOGUE: verified completed jobs | w=.5 is reconstructed. Unseen ratio15; no uniform benefit of physics assumed.

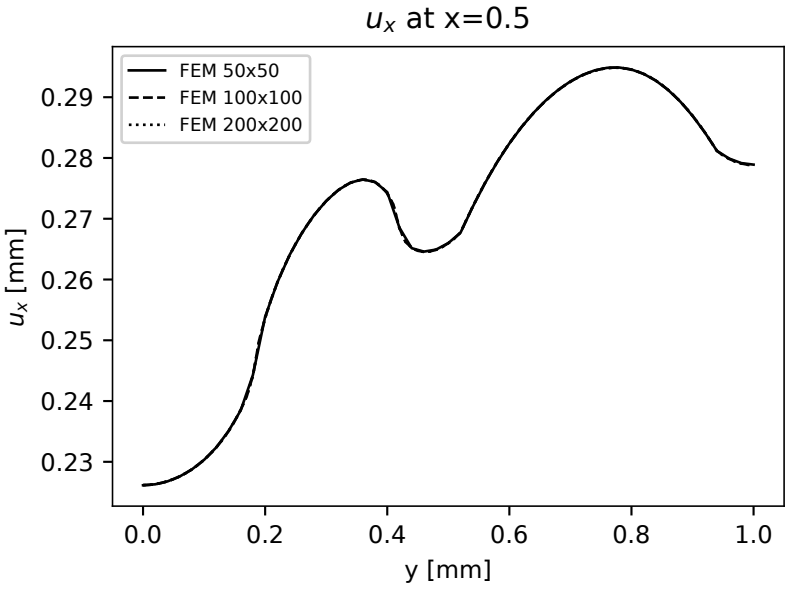
arch_l5_w40

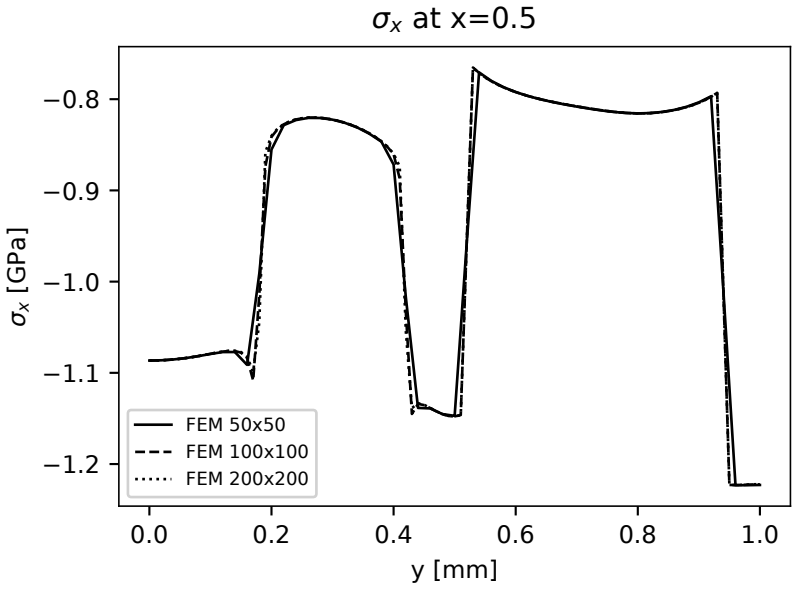


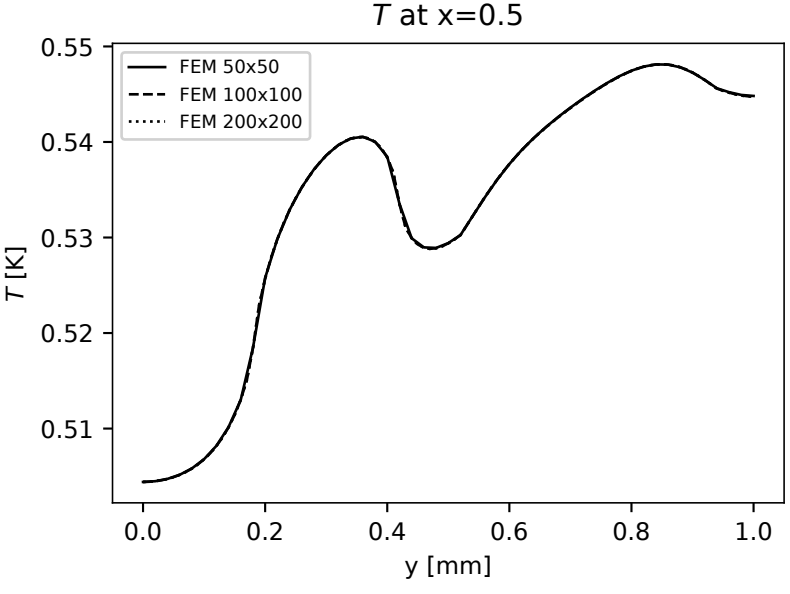
FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.

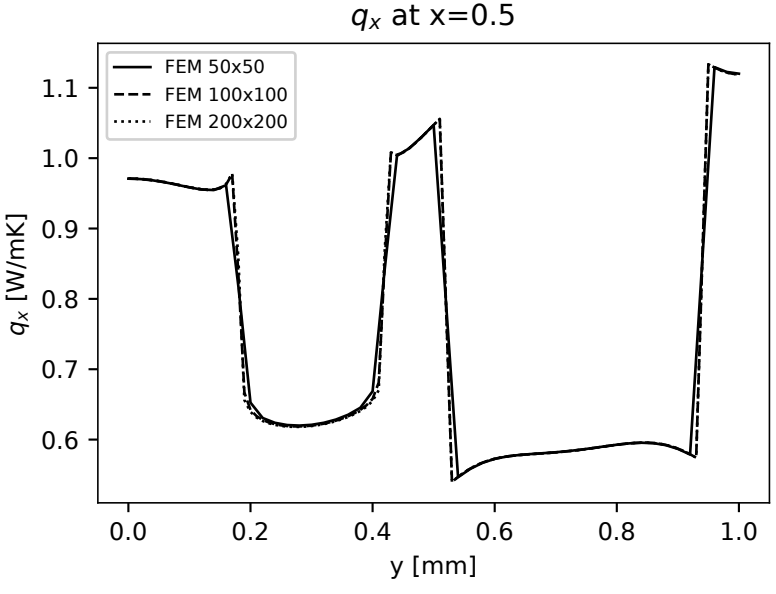


FULL CATALOGUE: verified completed jobs | Derived stress/flux; exact saved values. Printed limits where available; triangles disclose saturation.









FULL CATALOGUE: verified completed jobs | Independent FE mesh convergence; quantitative norms in JSON.